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SURVEY

A Comprehensive Review on ISAC for 6G: Enabling Technologies, Security, and AI/ML Perspectives

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ABSTRACT Integrated sensing and communication (ISAC) has received significant attention over the past five years as a key component of the 6G vision for next-generation communication systems. Starting with the fundamentals of ISAC, we present strategies focused on sensing-centric and communication-centric ISAC systems that satisfy both radar and communication objectives using the same physical hardware and waveform. The communication-centric ISAC focuses primarily on improving communication performance beyond 5G/6G cellular networks and Wi-Fi standards, while also performing additional processing required for detection. On the other hand, the sensing-centric ISAC prioritizes sensing and detection capabilities while incorporating communication functions in the most efficient way possible. In this survey, we present a comprehensive review of both ISAC systems along with new approaches for enabling technologies and highlight unique challenges and existing solutions that are already available. We explore the roles of innovative functions in enabling ISAC, including joint waveform design, edge computing, vehicular sensing and communications, and their practical applications. We then discuss 3GPP standardization efforts related to ISAC as a future direction for the 6G vision. We also outline possible future research directions, with a specific emphasis on security, artificial intelligence, and machine learning.

INDEX TERMS 6G, artificial intelligence, DFRC, edge computing, ISAC, mmWave, NOMA, radar, RIS.

I. INTRODUCTION

Integrated sensing and communication (ISAC) architecture has attracted great attention from both academia and industry, aiming to accomplish full integration of sensing and communication functions. The main goal of ISAC systems is to achieve spectral coexistence using a unified hardware platform, thereby reducing machinery costs and increasing spectral efficiency, while also addressing spectrum scarcity [1]. Research in ISAC domain focuses on optimizing the use of radio resources and hardware to ensure that the dual-purpose sensing and communication processes are adequately fulfilled. As the convergence of machinery tools and signal processing methodologies between communication and sensing domains increases, ISAC provides opportunities for improved efficiency in spectrum and energy usage, as well as reduction in computational complexity. Furthermore, the International Telecommunication Union

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(ITU) has explicitly outlined its expectation that ISAC will become a key component in next-generation wireless systems in International Mobile Telecommunications (IMT)-2030. Use cases for ISAC include assisted navigation, activity detection, movement tracking, environmental monitoring, artificial intelligence, and virtual reality applications [2].

With the adoption of the mmWave spectrum towards sixth-generation (6G) networks, the frequency bands used for radar and communications have started to overlap. This has motivated the development of dual-function devices that integrate radar and communication capabilities. ISAC has been known as joint sensing and communication (JSC) or dual-functional radar-communication (DFRC) in the literature [3] and was pioneered by McCormick et al. in 2017 [4]. Using the proposed far-field radiated emission design, communication and radar signals are generated as a function of time and spatial angle. Therefore, communication and radar signals can be propagated in two different directions using space division multiple access. The outcomes of this article have also been demonstrated experimentally.

Orthogonal frequency division multiplexing (OFDM) has been widely used in Wi-Fi, fourth-generation (4G), and fifth-generation (5G) systems. To overcome certain disadvantages of OFDM, orthogonal time-frequency space (OTFS) is proposed as an alternative waveform for 6G [5]. The popularity of OTFS is mainly due to its superiority in mitigating the disruptive effects of doubly-dispersive channels encountered in high-mobility communications. The delay-Doppler structure of OTFS makes it an ideal candidate for dual communication-radar applications [6], [7]. In OTFS modulation, delay and Doppler shifts can be estimated, as a result of the delay-Doppler structure of OTFS. Specifically, beam-sweep multiple-input multiple-output (MIMO) systems enable high-resolution velocity and range estimation, in addition to vehicle-to-vehicle, vehicle-to-network, and vehicle-to-everything (V2X) communications. OTFS offers greater robustness to high Doppler shifts compared to OFDM systems.

Sensing functionalities can also be very beneficial for mobile communications, especially during handover operations. While 5G systems employ beam-sweeping techniques to find the optimal direction for specific users, the importance of sensing functions is increasing in 6G. The 6G vision introduces numerous applications, trends, and technologies. Specifically, control, localization, and sensing are referred to as 3CLS and it is pointed out as a new trend for 6G in [8].

Another candidate hardware component of 6G systems is the reconfigurable intelligent surface (RIS) [5]. RIS allows the creation of visual line-of-sight (LoS) links where a direct LoS link is unavailable, thereby enhancing the system performance in terms of signal-to-noise ratio (SNR), outage probability, and secrecy capacity. Many studies address RIS and ISAC systems together [9], [10], [11], [12], [13], [14]. These works demonstrate that RIS-aided ISAC systems perform better, both in terms of radar detection and communication, compared to systems without RIS.

Artificial intelligence (AI) techniques are used by ISAC systems to enhance their sensing and communication capabilities. AI-driven approaches in ISAC systems offer numerous advantages including improved accuracy in target detection and channel estimation, real-time adaptability to dynamic environments, and optimized resource allocation [15], [16], [17], [18]. Machine learning (ML) techniques allow ISAC systems to predict and adapt to changes in the environment that enhance the overall efficiency and reliability of communication networks. Additionally, AI algorithms enable more efficient beamforming and signal processing, which leads to increased data rates and reduced latency. These advancements make AI an indispensable component in the development of next-generation ISAC communication systems.

As AI/ML techniques become more prominent, the need for computational power rises significantly. Mobile edge computing is a suitable paradigm that can address this challenge. Edge servers with sufficiently high computing resources can be deployed either statically in the infrastructure such as base stations (BSs) or roadside units (RSUs), or can be placed on components such as vehicles and unmanned aerial vehicles (UAV). Computation offloading can serve as a facilitator for ISAC, but it can also use ISAC to infer certain parameters such as user velocity and direction, which facilitate to improve offloading performance, typically in terms of delay.

ISAC-related standardization and regulatory matters are managed by the 3rd Generation Partnership Project (3GPP), which defines the use cases and requirements for ISAC services to address various applications in enhancing 6G wireless networks. In 3GPP's framework for 6G mobile systems, ISAC integrates sensing and communication capabilities within the same wireless communication system and framework. In general, 3GPP addresses ISAC-associated scenarios of communication-assisted models, where sensing information is integrated within communication channels or environments [19], [20]. This new architecture planned by 3GPP is expected to enhance communication services within 6G wireless systems while enabling new services and use cases. ISAC-derived sensing information can be used to assist the end-user services for autonomous driving, V2X, AI/ML solutions, and a broad range of new functionalities, such as object detection, tracking and activity recognition, and high accuracy localization. Finally, ISAC presents applications offering trajectory tracing, collision avoidance, and traffic management as shown in Fig. 1. Handling of the above ISAC-associated enhanced services and applications is standardized, and the ISAC protocols are regulated by the 3GPP technical specification group [21], [22], [23].

Research in ISAC has also recently gained significant attention within the V2X framework, as mobile communication systems with radar functionality for vehicles promise high-throughput and high-accuracy communication potentials. The high mobility of the vehicles, together with the ultra-low latency requirements of traffic applications, poses critical challenges in vehicle tracking. For instance,

robust and high-resolution vehicle sensing/localization, anti-collision detection, precise location estimation, and accuracy are vital specifications for autonomous vehicle applications. The ISAC is typically motivated by the vehicular networks. ISAC systems should optimize the utilization of the spectral resources and enable the effective cooperation between communication and sensing functionalities for the above-mentioned scenarios [24], [25], [26]. The aforementioned technologies constitute both use-cases and enablers for ISAC. Each has its own unique effect on potential ISAC designs and systems.

ISAC is strongly anticipated to combine advanced sensing capabilities with communication technologies in 6G, promising transformative impacts across various domains. In smart cities, for instance, ISAC can enhance real-time monitoring and adaptive infrastructure for enhanced and effective urban management. For automated industries, it enables seamless machine communication, predictive maintenance, and optimized supply chains. In agriculture, ISAC supports precision farming through data-driven decision-making and efficient resource management. In healthcare, it improves wearable devices, telemedicine, and personalized patient care. Overall, ISAC is poised to form more interconnected, efficient, and responsive systems in everyday applications.

Several pioneering tutorial and survey papers have discussed ISAC-related research. The review in [27] examines the industrial progress and standardization activities related to ISAC, including ongoing and potential use cases. Several performance trade-offs between sensing and communication are analyzed, spanning from information theoretical limits to physical layer performance trade-offs and cross-layer design trade-offs. ISAC waveform design is discussed in terms of signal processing aspects in the same work. Potential integration of ISAC with other emerging communication technologies is identified, along with the positive impacts on future wireless networks. In [28], the authors review the literature on ISAC signals from the perspective of signal design, signal processing, and signal optimization. They discuss radar signal processing methods for ISAC signals, including the channel information matrix method, line spectrum estimation, and superresolution techniques. Subsequently, based on the signal optimization approach, peak-to-average power ratio (PAPR) optimization, interference management, and adaptive signal optimization for ISAC signals are summarized for ISAC signals in 5G-advanced and 6G wireless communication systems.

Another review of the recent advances in ISAC systems is provided in [29], with a particular focus on physical layer system design, networking, and ISAC applications. The review also addresses open challenges, including information theoretic limits, synchronization and phase offset issues, cross-layer resource management techniques, transmission protocols, and security and privacy issues of ISAC systems. The overview article in [30] identifies several challenges, applications, and opportunities for ISAC in the internet of things (IoT). A survey in [31] explores ML-enhanced ISAC systems, focusing on their architectures, algorithms, and applications. This comprehensive paper provides insights into how ML techniques can enhance ISAC systems and improve their performance.

The works mentioned above provide an overview of ISAC's key technologies, including scenarios and requirements. However, many other aspects of ISAC systems, such as vehicular communication and edge computing scenarios, RIS-aided ISAC, NOMA in ISAC systems, ISAC in UAV applications, security issues, AI-ML trends in ISAC, as well as the 3GPP standardization process of ISAC yet remain underexplored.

A. KEY CONTRIBUTIONS

To fill in the gap on a broad perspective into the enabling and enhancing technologies and their interactions within ISAC, we provide a comprehensive review of ISAC systems from a number of perspectives in terms of communication, sensing, edge computing, security, and AI. Fig. 1 illustrates prospective implementation scenarios for ISAC systems. To improve the readability of the survey, a list of acronyms is given in Table 1. This survey primarily focuses on ISAC systems within the context of the technologies given in Fig. 2. Several survey studies have examined ISAC technologies from different perspectives in recent years, often focusing on specific technical aspects such as waveform design, resource allocation, or sensing-assisted communication. However, while these studies have provided valuable insights into either sensing-centric or communication-centric approaches, they generally offer limited discussions on recent practical developments in an integrated manner. Unlike previous surveys, this study presents a more comprehensive and structured overview by first classifying ISAC studies into communication-centric and sensing-centric approaches. The survey also discusses the practical applications of ISAC systems, the challenges encountered, and their role in 3GPP standards. Particularly, differing from the preceding related surveys and studies, we provide an in-depth review of the integration and co-existence of the following technical concepts with ISAC:

- Vehicular/Edge computing,
- RIS-aided ISAC systems,
- Security issues,
- AI/ML perspectives.

Table 2 summarizes a representative selection of the studies in the literature under different clusters, including communication and radar objectives, type of waveforms, vehicular applications, RIS-aided systems, and the use of AI. In this way, the study aims to provide a broad perspective on both theoretical and practical aspects of ISAC and contribute to identifying open issues and potential research directions.

B. ORGANIZATION OF THE PAPER

The organization of the survey is outlined in Fig. 3. Section II introduces the fundamentals of ISAC, focusing on communication-centric and sensing-centric solutions.

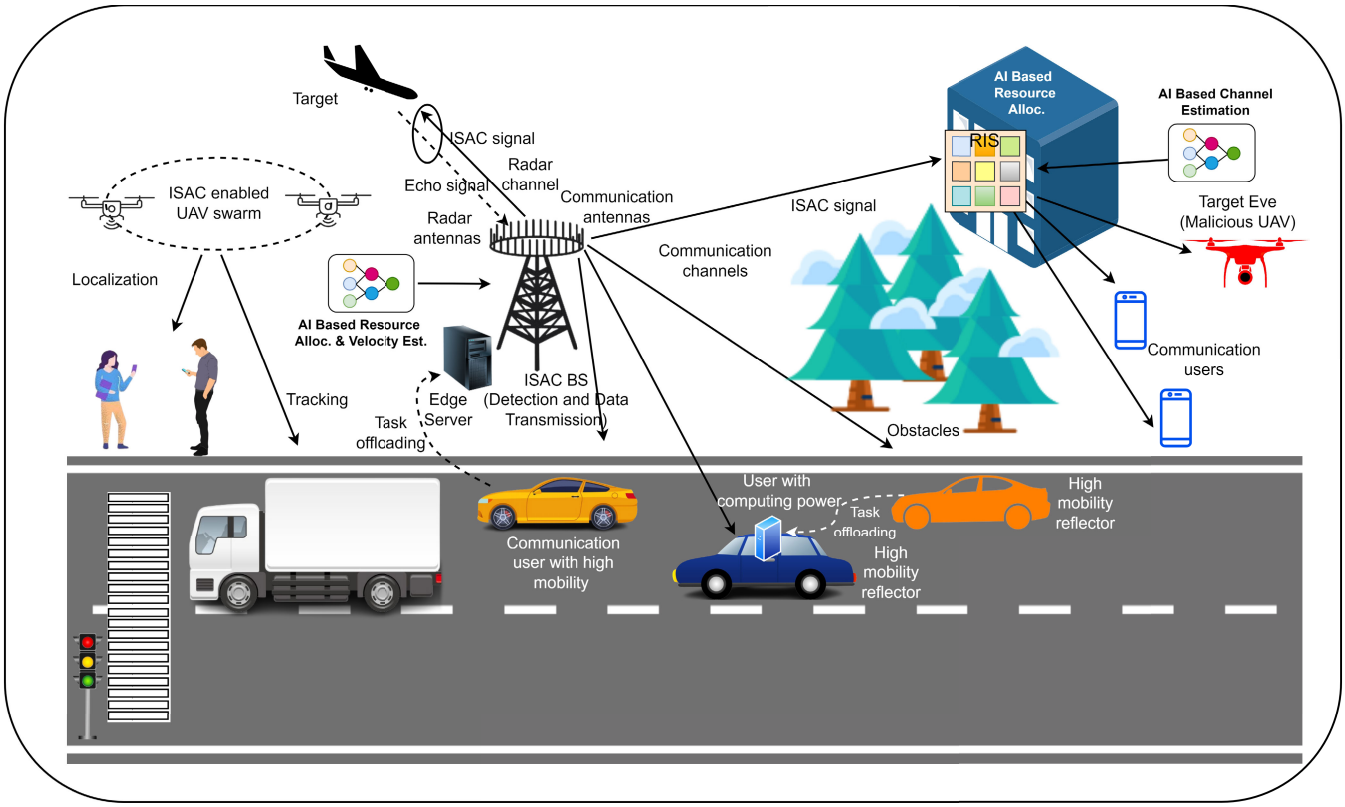


FIGURE 1. Potential application scenarios for ISAC systems. ISAC can support communication systems with localization, target detection, and tracking capabilities. These systems can be further enhanced with RIS and/or UAV-swarm technologies. ISAC may induce unique security challenges as the targets to be sensed could be inadvertently exposed to communication data.

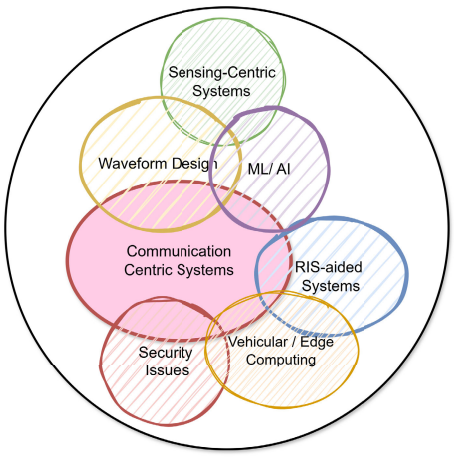


FIGURE 2. Different perspectives of ISAC systems.

Section III presents new approaches for enabling ISAC systems including joint waveform design and signal processing schemes, edge computing, vehicular communications, and practical communication in ISAC architectures. Section IV discusses 3GPP standardization on ISAC systems. Section V addresses security issues, emerging challenges, and their corresponding solutions. Moreover, AI and ML for ISAC systems are reviewed in Section VI. Finally, Section VII provides the conclusion.

II. ISAC FUNDAMENTALS

ISAC for moving target imaging and tracking is becoming increasingly important. ISAC systems perform dual functions by simultaneously achieving wireless communication and radar sensing through a single transmitted waveform. The ISAC design enables communications and radar sensing by sharing the same resources, such as frequency bands and hardware, thus increasing spectrum efficiency and reducing cost, which brings some trade-offs. Studies have shown that there may be a decrease in the performance of these systems while performing both functions together, compared to the case where two functions are implemented separately. In this section, we delve into communication-centric and sensing-centric ISAC approaches and explain the relevant literature for both of them in detail.

A. COMMUNICATION-CENTRIC ISAC APPROACH

The communication-centric ISAC approach offers detection capabilities in wireless networks, including beyond 5G (B5G)/6G cellular and Wi-Fi networks [32], [33]. The communication-centric ISAC system design primarily focuses on achieving high-throughput wireless communication while enabling sensing functionalities, as illustrated in Fig. 4. Sensing can be performed on the communication waveform reflected or scattered by targets during sensing

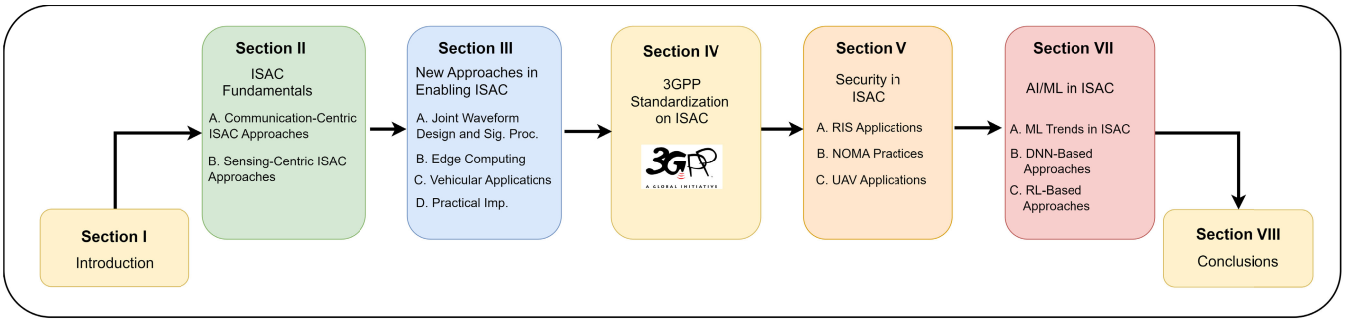


FIGURE 3. Organization of the survey.

periods. Targets can be identified and tracked by observing the changes in the propagation environment. As the achievable rate and bit error rate (BER) are key performance metrics in such a system, the sensing performance cannot always be guaranteed. Since the system model is designed from a communication-centric perspective, additional processing can be required for detection [34]. Communication-centric ISAC sensing architecture can be considered based on the scenario and system model. The system model can be classified into mono-static or bi/multi-static sensing configurations. In a mono-static sensing configuration, the transmitter and receiver are at the same location, as shown in Fig. 4. That is, firstly, the antenna(s) is (are) used to transmit the sensing signal (as a radar pulse) and the returning echo of the sensing signal is then received by the same antenna(s) (or the transmit/receive antennas are separate but at the same location). In bi/multi-static sensing configuration, the receive antenna(s) is (are) not the same as the transmit antenna(s), and the transmit and receive antenna(s) are at different locations, as illustrated in Fig. 5. Both configurations have their own advantages and disadvantages and different challenges emerge in each configuration [35].

A communication network includes many nodes that together serve up coverage area. Therefore, the infrastructure of a bi/multi-static sensing network with ISAC technology is mostly already available or suitable. Multiple receivers can be used to receive echoes from a single or multiple nodes. Multiple sensors are already available that can detect reflections from various directions using one or more transmission nodes. Network nodes are currently connected to the core network via the backhaul, enabling essential interconnections to share sensing pulse configurations and potentially locally (pre-)processed sensor echoes with other nodes or a central sensing node [36]. The distance between the transmitter and receiver stations of a bi/multi-static radar can be large, and in most of them the azimuth angle differences can also be large. Even if the reflective target has no or very little energy reflected in the direction of a mono-static radar due to its geometry designed with the so-called stealth technology, it may be possible to receive these echoes with a bi/multi-static sensing mechanism [37].

In the multi-static scenario, a synchronization framework is required between nodes to coordinate transmissions, both to keep track of when transmitted signals should be listened to and to hold inter-node interference under control. Node-specific reference signals that identify the transmitter node are also needed, and these signals can already exist in the context of the communication network. In addition, the clock asynchronism problem encountered in the bi/multi-static detection topology can cause timing offset (TMO) and carrier frequency offset (CFO). Moreover, the reflective surface of the target is not the same for each sensor antenna. This requires the use of much more complex target models in sensing data processing. Furthermore, if communication-assisted sensing is planned to achieve higher performance in a simple ISAC bi-static scenario, the channel coefficient for differently positioned sensor receivers/antennas and communication signal need to be known (or estimated) while the superimposed communication and sensing signal is transmitted [38].

A mono-static setup is also commonly used in traditional radar applications. In this classic scenario, a mobile radar platform (e.g., aircraft, ship, car, etc.) uses radar technology for tracking civilian targets, collision avoidance, and other types of mapping of the environment. A mono-static setup can also be applied in communication networks, but it introduces some requirements to the radio, such as the following [36].

- A radio in a mono-static setup must have a self-interference (Tx-to-Rx) cancellation mechanism to prevent echoes from very close objects from being received back and saturating the receiver.
- If the pulse duration is short or if echoes arrive from a distance away, this self interference block may not be needed.
- If Tx/Rx are performed with the same antenna, the Tx-to-Rx switching time should still be fast enough to catch the first incoming echoes. However, communication systems are generally not equipped with very fast switches.

1) MONO-STATIC SENSING CONFIGURATION FOR COMMUNICATION-CENTRIC ISAC APPROACH

Within the scope of mono-static sensing configuration, the BS is used for radar operations while transmitting the downlink modulated signal at the same time, as in [39]. Therefore, the BS is concurrently employed as a mono-static combined radar

TABLE 1. The abbreviation list.

Abbreviation	Definition
3GPP	3rd Generation Partnership Project
AF	Ambiguity function
AFDM	Affine frequency division multiplexing
AI	Artificial intelligence
AN	Artificial noise
BER	Bit error rate
BS	Base station
CAS	Communication aided sensing
CFO	Carrier frequency offset
CNN	Convolutional neural network
CR	Communication rate
CRLB	Cramér–Rao lower bound
CU	Communication user
CSI	Channel state information
DDPG	Deep deterministic policy gradient
DFRC	Dual-functional radar-communication
DL	Deep learning
DNN	Deep neural network
DRL	Deep reinforcement learning
DT	Digital twin
EE	Energy efficiency
FEEL	Federated edge learning
FL	Federated learning
FP	Fraction programming
GAI	Generative artificial intelligence
GAN	Generative adversarial network
GLRT	Generalized likelihood ratio test
HCL-Net	Convolutional long short-term memory network
HFS	Hopping frequency sequence
HBF	Hybrid beamforming
IM	Index modulation
IOS	Intelligent omni-surface
ISAC	Integrated sensing and communication
JSC	Joint sensing and communication
LFM	Linear-frequency modulation
LMMSE	Linear minimum mean square error
LSTM	Long short-term memory
LTE	Long term evolution
MEC	Mobile edge computing
MIMO	Multiple-input multiple-output
MISO	Multiple-input single-output
ML	Machine learning
NOMA	Non-orthogonal multiple access
NR	New radio
OCDM	Orthogonal chirp division multiplexing
OFDM	Orthogonal frequency division multiplexing
OMA	Orthogonal multiple access
OTFS	Orthogonal time-frequency space
PAPR	Peak-to-average power ratio
PLS	Physical layer security
PRS	Positioning reference signal
PSO	Particle swarm optimization
RIS	Reconfigurable intelligent surface
RMSE	Root mean squared error
RSU	Road side unit
QoS	Quality of service
SAC	Sensing and communication
SAR	Synthetic aperture radar
SDP	Semi-definite programming
SI	Self-interference
SR	Sensing rate
SNR	Signal-to-noise ratio
SINR	Signal-to-interference-plus-noise ratio
STAR	Simultaneously transmitting and reflecting
STO	Sampling time offset
TCM	Transmit covariance matrix
TMO	Timing offset
UAV	Unmanned aerial vehicle
UE	User equipment
ULA	Uniform linear array
V2I	Vehicle-to-infrastructure
V2X	Vehicle-to-everything

transceiver and communication transmitter with a common waveform in the system model. The Doppler shift caused by the relative mobility between the transmitter and receiver results in the phase of the received symbol being different from the phase of the transmitted symbol at the BS. The range and velocity of the target can be determined by estimating the delay and the Doppler shifts with minimum error variance and bias. The Cramér-Rao lower bounds (CRLBs) for range and velocity define the theoretical minimum achievable error variances. According to [39], waveform optimization is based on the assumption that the OFDM transmit frame contains empty subcarriers, indicating that it is not fully loaded. The optimized frequency-domain samples are used to align with empty subcarriers to enhance detection performance. Therefore, if the total transmit power is limited, a portion of the subcarrier power is reallocated to the radar subcarriers. This effectively controls the performance balance (fairness) of these two subsystems. Using the outlined optimization techniques, range and velocity estimates, as well as the transmission waveform's PAPR, are effectively controlled and minimized. This approach does not affect the communication receiver design, as the radar subcarriers can be discarded when detecting the communication data.

Another mono-static OFDM ISAC system, which estimates target parameters using backscattered signals while simultaneously communicating with an OFDM receiver, has been studied in [50]. The goal is to achieve an optimal performance trade-off between the communication and radar operations by adjusting subcarrier powers on the time-frequency grid. To achieve this, a radar-optimal waveform design has been developed to minimize errors in delay-Doppler estimation. This design considers an integrated sidelobe level constraint in the delay-Doppler uncertainty domain and optimizes waveform design to optimize communication performance while adhering to radar similarity constraints. The presence of a low-rate feed-forward channel (carrying transmission waveform control information) between the ISAC transceiver and the communications receiver is assumed. The communication metric is defined as a function of both subcarrier powers and transmitted control information. Then, this communication metric is optimized by employing a joint waveform and control signaling optimization (JWCSO) strategy to minimize a linear minimum mean square error (LMMSE).

The proposed JWCSO approach presents near-optimal radar communication trade-off performance by reaching the CRLB with only limited feed-forward information. Compared with previous studies, the study in [70] focuses only on minimizing the PAPR. As is well-known, high PAPR can cause nonlinear distortion in transmit signals, leading to reduction in radar/communication performance due to limited linear region of low-cost power amplifiers. The solution is demonstrated by using a weighted objective function in both communication and radar performance measurements for designing a MIMO-OFDM waveform with low PAPR in ISAC under power and PAPR constraints.

TABLE 2. Comparison of selected research papers on ISAC.

Scenario	References	System Model / Technology	Contribution in Communication Part	Contribution in Sensing Part	AI
Communication-Centric	[32]	mmWave massive MIMO	Pilot overhead reduction	Estimation of angles, delays, and the Doppler frequencies	
	[39]	Mono-static radar	Capacity	Target range and velocity estimation	
	[40]	Bi-static radar	Satisfy communication requirement	Maximize radar SINR	
	[41]	Multi-static radar	Communication spectral efficiency	Sensing spectral efficiency	
	[42]	MIMO	Communication rate	CRLB minimization	
	[43]	NOMA	Maximize sum rate	Maximize sensing power	
	[44]	mmWave beam tracking	Communication throughput, latency	Target detection	
Sensing-Centric	[45]	ISAC system	Achievable bit rate, BER	Transmit messages in the form of IM	
	[46]	MIMO	Outage probability, communication rate, diversity order	Sensing rate	
	[47]	Full-duplex (FD) system	Communication rate, probability of detection	Target sensing	
	[48]	Beamforming design	RMSE	Target sensing	
	[49]	Trade-off between communication and sensing	Communication rate	Target sensing	
Waveform Design	[50]	OFDM ISAC	Capacity, RMSE	Target estimation accuracy	
	[51]	MIMO ISAC	Achievable rate, MSE, SER	Target detection	
	[52]	Delay-Doppler OTFS	BER, capacity, MSE	Multi-target sensing	
	[53]	Discrete Fourier transform spread OFDM	BER, RMSE, achievable rate	Target range and velocity estimation	✓
	[54]	THz OTFS	BER, RMSE	Multiple target estimation	
Vehicular / Edge Computing	[55]	Sensing aided OTFS	BER, NMSE, False alarm probability	Target detection	
	[56]	Beamforming design for V2I	Achievable rate, RMSE	Target detection	
	[57]	ISAC for vehicular networks	RMSE	Target detection	
	[58]	MCW for sensing, QAM for communications	Optimal transmit power	Human motion recognition	✓
	[59]	Multiple ISAC devices using short packets to offload sensing data to an edge server	Energy efficiency, latency	Guaranteed mutual information between ISAC device and target	
Security	[60]	Vehicular communications	Concealing confidential information	Beam tracking	
	[61]	Multi-user downlink with a multi-antenna BS	A trade-off between sensing and secrecy performances	Sensing malicious targets	
	[62]	Multi-user downlink with ISAC BS	Joint optimization of communication and sensing resources	Acquisition of CSI of the potential eavesdroppers	
	[63]	Multi-user MISO with a malicious UAV	Maximize the attainable secrecy rate via an active RIS	Tracking the UAV	
	[64]	ISAC NOMA	Maximize the sum secrecy rate while satisfying a definite level of sensing performance	Sensing position of a target, which acts as a possible eavesdropper	
RIS-Aided	[9]	Beamforming and reflection design	Radar SNR maximization	Enhancing sensing performance	
	[10]	mmWave system	Communication SINR	Target sensing	✓
	[11]	Beamforming design	Maximize the radar information	Vehicle target detection	
	[12]	Secrecy rate optimization	Secrecy rate, MSE	Target tracking	✓
	[13]	Channel estimation	NMSE	Target sensing	✓
AI / ML	[65]	Passive sensing and communication	Signal detection and channel reconstruction modules for BER and SER	Range and velocity estimation	✓
	[66]	6G wireless networks	Mitigating the mutual interference via joint sensing transmit waveform and communication receive beamforming design	Same as communication contribution	✓
	[67]	Vehicular networks	Predictive beamforming design (Achievable rate)	-	✓
	[68]	RIS aided ISAC (ISAC BS communicates with multiple IoT devices)	-	Location and velocity estimation	✓
	[69]	Fluid antenna system (FAS) and MIMO	FAS in the downlink for multiuser MIMO system	Channel state information	✓

Along with the OFDM waveform, [71] summarizes and compares OTFS, orthogonal code division multiplexing (OCDM), and affine frequency division multiplexing (AFDM) waveforms for mono-static sensing ISAC systems. AFDM can provide lower computational complexity for channel estimation overhead. Decoupling

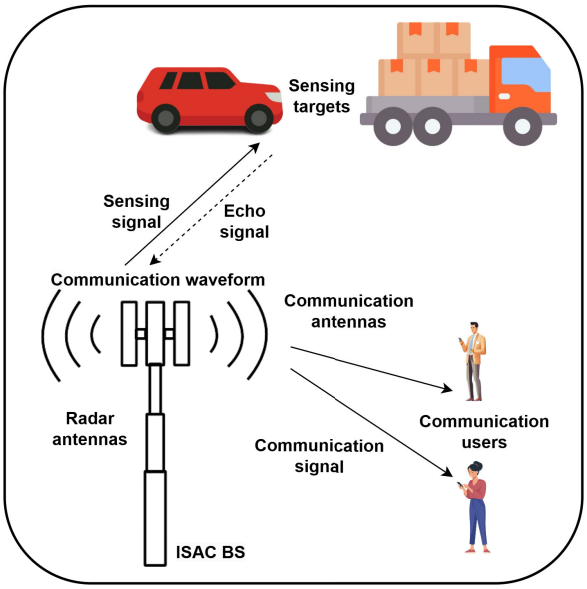


FIGURE 4. Illustration of the communication-centric ISAC scenario. The ISAC BS with multiple antennas provides communication and sensing functionalities simultaneously in the mono-static communication-centric ISAC. The sensing functionality can address road safety problems.

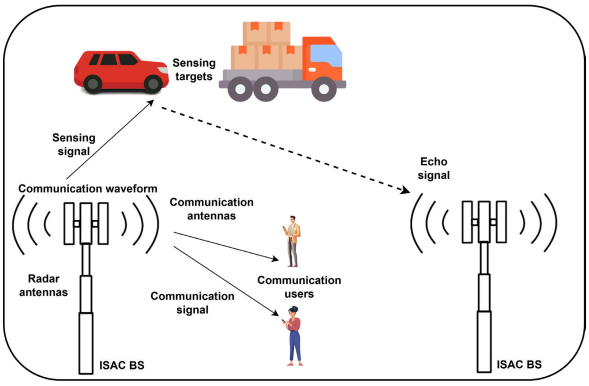


FIGURE 5. Bi-static sensing configuration.

delay and Doppler shift with AFDM-based waveform is also possible even under high mobility conditions. Derived or combined waveforms can also be considered. The study in [72] addresses the delay-Doppler coupling effect in single-frequency linear-frequency modulation (LFM) by using triangular LFM, to decouple range and Doppler ambiguity. Furthermore, the combination of triangular FM and OFDM, as a communication-centric waveform, outperforms OFDM-based mono-static sensing in the ISAC scheme.

Reference [73] proposes a fluid antenna-assisted ISAC system, in which a BS detects a target while transmitting signals to a communication user (CU). Unlike traditional ISAC systems with fixed-position antennas, fluid antennas are equipped at the BS and CU and are dynamically adjustable within a certain area. By jointly optimizing the transmission beamforming and the positions of fluid antennas at the BS and CU, the downlink communication speed is maximized while satisfying the sensing beam pattern gain and transmit power requirements of the BS. Another study investigates a fluid antenna-assisted ISAC system with combined antenna position optimization and waveform design [74]. With the help of fluid antennas, sum-rate maximization, improved of sensing performance, and alleviation of design computational complexity in fluid antenna-assisted ISAC systems are presented. Another study investigates the movable antenna assisted multi-user ISAC system to improve both the sensing and communication performances of the BS and communication users [75]. CRLB is used as the performance metric of detection, so a joint beamforming design and movable antennas location optimization problem is formulated to minimize the CRLB.

Regarding hardware implementation, an ISAC system using the 5G NR waveform has been developed on the ZCU208 RF system on chip (RFSoC) platform [76]. Data visualization is performed by using the PYNQ platform. Both the communication and sensing aspects of the design have been verified to work in field tests. Reference [77] demonstrates the benefits of a co-designed system by comparing performance with and without cooperation between radar and communications. A classical interference reduction approach called temporal reduction is implemented to enable efficient joint radar communications on the Xilinx ZCU102 FPGA evaluation platform. Performance results of the practical hardware design are presented, demonstrate lower resource utilization and shorter latency compared to the software process.

2) BI/MULTI-STATIC SENSING CONFIGURATION FOR COMMUNICATION-CENTRIC ISAC SYSTEMS

In bi-static radar, a common high-precision clock signal synchronization is generally required, typically achieved through fiber optic links or GPS. However, in current communication systems, this is often not possible between the BS and the user equipment (UE). In this case, the clock asynchronism problem encountered can create TMO and CFO. Both TMO and CFO gradually change over time due to clock instability. In communication, TMO can be overcome by channel estimations, and CFO can be estimated and compensated. In contrast, TMO and CFO produce uncertainty in measurement and accuracy degradation during detection. TMO can directly produce timing uncertainty and thus range uncertainty, and CFO can produce Doppler estimation uncertainty and thus velocity uncertainty. In [78], three solutions are presented to address this problem. The first is the use of a global reference clock for outdoor terminals. It requires satellite visibility and is limited to outdoor scenarios; solutions that strike a good balance between synchronization speed and accuracy have not yet been developed. Secondly, single node-based techniques require multiple channels with multiple cross antennas and

the use of locked clocks on multiple receiver channels. Thirdly, network-based techniques require addressing difficult target association problems when applying triangulation techniques. The authors state that all three categories of methods can be combined.

For bi-static sensing architectures with high data rate requirements, both BS and CU have multiple antennas and the communication system is designed as a MIMO system in [42]. In this scenario, only BS senses the extended target simultaneously within the mono-static and bi-static architecture. CRLB serves as a basic performance metric for radar estimation, defining the lower bound of variance by unbiased estimators. Moreover, channel capacity serves as a basic performance metric for wireless communication that captures the communication rate upper bound by any practical modulation and coding scheme. The fundamental trade-off between the CRLB for sensing and the data rate for communication is investigated by featuring the Pareto bound of the achievable CRLB-rate (C-R) region. The C-R region boundary between the CRLB minimization and rate maximization points is considered. For this purpose, a new MIMO rate maximization problem is formulated by optimizing the transmission covariance matrix at the BS by subjecting it to a new form of maximum CRLB constraint along with the maximum transmit power constraint.

Another bi-static ISAC MIMO architecture study in [40] proposes a joint transmit and receive beamforming design using a hybrid beamforming structure in a bi-static radar sensing and multi-user MIMO (MU-MIMO) configuration. The study focuses on tracking targets with a optimized directional beamforming design. To deal with the CFO and sampling time offset (STO) problem encountered in bi-static ISAC detection topology, the LoS path signal is considered as a virtual target. The radar SINR value for the virtual target is designed to be larger than a threshold value to estimate and eliminate the effect of CFO and STO. Hence, the transmitter and sensing receiver beamformers are jointly designed to maximize the sensing SINR while meeting the SINR constraints of both communication users and the virtual radar target. To solve this problem, a hierarchical solution algorithm is proposed where a combined fully digital beamformer is first optimized for the transmitter/detection-receiver, and then the analog and digital beamformers are derived from the fully digital beamformer for hybrid beamforming. It is understood that the peaks in the transmitted and received beam patterns match the user and target directions.

Multi-static sensing ISAC systems can provide enhanced sensing performance. The authors focus on sensing coverage for cellular ISAC systems in [79], aiming to cover a wide area within the sensing coverage region while simultaneously serving a group of communication UEs for cellular ISAC systems. Radar sensing signal processing operations and communication system models are run in a general multi-static cellular ISAC scenario with coordinated multi-point joint transmission. Beamforming optimization is also implemented in the multi-static sensing ISAC system to maximize the sensing probability of the desired coverage area. As a benchmark comparison, only the communication beamforming design is also included, which minimizes the transmission power based on the signal-to-interference plus noise ratio (SINR) need for each UE, without including the sensing coverage need.

Another work in [41] focuses on multi-static sensing for ISAC in the availability of clutter. ISAC access points send downlink signals to UEs and, if desired, direct a beam toward the target in a multi-static target detection scenario. Two ISAC configurations are considered in beam steering: employing real communication beams for sensing or using other sensing beams. A maximum a'posteriori ratio test detector is proposed for target sensing in the presence of clutter. The concept of sensing spectral efficiency is introduced in ISAC as a metric that captures the effect of resource utilization. A power allocation algorithm is also developed to maximize SINR and ensure compliance with minimum communication requirements are met. A multi-static MIMO ISAC system in cellular networks is proposed in [80]. It can use multiple BSs to realize collaborative sensing over a wide area. In this system, the communication reference signals sent from the central BS are received by the neighboring BSs. Then, joint data processing is performed to estimate the location and speed of the targets, and superior detection accuracy is achieved compared to mono-static systems.

Bi-static inverse synthetic aperture radar (Bi-ISAR) imaging technique for non-cooperative moving targets using OFDM signals for ISAC is proposed in [81]. In the OFDM Bi-ISAR system model, the keystone transform is employed for non-cooperative targets to adjust migration along the range cell and eliminate phase errors on phase compensation.

3) BEAMFORMING DESIGN FOR COMMUNICATION-CENTRIC ISAC SYSTEMS

Communication or sensing efficiency is increased with different beamforming designs. For this reason, array antenna beamforming has an important place in communication-centric ISAC system models, especially at BSs [42], [43], [78]. In this context, several optimization studies on beamforming are relevant. NOMA communication signal is used for simultaneous target detection in [43]. The goal of this study is to maximize the weighted sum of communication performance and efficient detection power while contenting each user's minimum communication capacity and radar-specific needs. A beamforming optimization problem is defined to acquire the communication-sensing trade-off, which is addressed by a double-layer penalty-based algorithm. Energy efficiency (EE) optimization in beamforming design for an ISAC scenario is explored in [82] considering both communication-centric EE and sensing-centric EE, and their trade-off. The goal is to maximize system EE while considering constraints such as maximum transmit power, minimum desired SINR for communication,

and maximum estimation performance for target detection. In addition, the Pareto bound of communication-centric EE and sensing-centric EE is examined and their trade-offs are characterized.

In [33], an optimal transmit beamforming method for ISAC systems is presented. A single BS with a uniform linear array (ULA) transmits joint information-carrying and sensing radar signals to simultaneously optimize downlink multi-user communication and multi-target radar detection. This study shows that the proposed minimum-weighted beam pattern gain maximization design outperforms the beam pattern matching design. Moreover, Type-I and Type-II receivers are discussed, which either have or do not have the ability to cancel interference from previously known specific radar signals, respectively. Further, it has been shown that Type-II receivers provide better detection performance than Type-I receivers or other conventional designs. In another study, a robust energy-efficient beamforming design for full-duplex ISAC communication systems is investigated [83]. The average EE and average spectral efficiency are compared for the proposed EE maximization algorithm and the benchmark spectral efficiency maximization algorithm.

The authors in [84] propose an ISAC model that incorporates beamforming design to sense dynamic targets within a cluttered environment while ensuring communication quality for users. The system forms multiple communication beams to interact with users and one detection beam that scans the entire area. In order to minimize the interference of sensing beam in existing communication systems, after dividing the service area into sensing beam for detection sector and communication beam for sensing sector, and focusing on the beamforming design and power allocation optimization strategies for each type of sector. Dynamic target sensing and angle estimation are realized by angle-Doppler spectrum estimation and joint detection on multiple subcarriers, while range and speed estimation are realized by extended subspace algorithm. Beamforming also offers benefits in ISAC systems where mobile vehicles are present. The scenario of a UAV-aided ISAC system is discussed in [85]. Specifically, it is shown that the jittering effect caused by UAV movement and attitude variation can decrease communication performance by altering beam direction. As a solution to this problem, a high-performance ISAC-based channel estimation technique is proposed in the study.

The authors in [86] discuss the self-interference (SI) problem, where the transmitted ISAC signal leaks to the receiver side of the BS. A new nonlinear SI cancellation for a full-duplex ISAC scenario in V2X networks is proposed to mitigate SI. First, an improved accuracy tap channel model is derived to characterize the nonlinear SI by considering the fractional delay. The proposed scheme can be implemented with or without channel state information (CSI) of SI, where the reconstruction coefficients are obtained by reflecting the coefficients from the path signal model or updating the previous reconstruction coefficients, respectively. Downlink

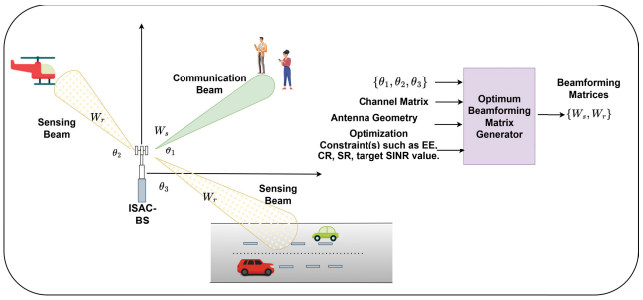


FIGURE 6. Optimum beamforming matrix generation concerning user and target locations, constraint values.

multi-user communication and multi-target radar detection are conducted using a single BS with N_r antennas for radar and N_t antennas for communication in [87]. The radar signal covariance matrix and the communication physical precoder are jointly designed to achieve a proper beam pattern for sensing and maximizing the SINR of the downlink transmission. This approach reduces the computational complexity at the cost of a minor loss in system performance. It has been stated that the proposed method results in relatively higher sidelobe loss. Fig. 6 illustrates a scenario where beamforming matrices for radar $\mathbf{W}_r$ and communication $\mathbf{W}_s$. Moreover, several inputs are given to generate $\mathbf{W}_r$ and $\mathbf{W}_s$.

4) VERY HIGH FREQUENCIES FOR COMMUNICATION-CENTRIC ISAC SYSTEMS

Communication-centric ISAC systems can also operate at very high frequencies. Especially, mmWave and THz bands offer high communication speed and high resolution in sensing [43], [88]. By utilizing the high angular resolution of mmWave massive MIMO, the shapes of targets can be obviously specified and imaged by utilizing radar echoes from different directions. Compressed sampling (CS), also known as compressed sensing, can recover signals from mitigated measurements by exploiting the intrinsic sparsity. The motivation behind the radar applications of CS approaches is the inherent "sparse" nature found in many radar problems (e.g., limited number of relevant targets or compressibility of radar images). In communication theory and practice, mmWave mMIMO with hybrid beamforming (HBF) architecture has been taken into account as a key enabler for 5G/B5G. The ISAC systems based on mmWave massive MIMO with HBF architecture and the use of compressed sensing methods are presented in [32]. By recommending an ISAC frame design and waveform scheme adapted for HBF architecture and CS processing, the realization of radar sensing with channel estimation in conventional cellular communication is studied. The angular resolution of radar detection, the estimation of delay and Doppler frequencies are improved, and the pilot load is minimized.

Infrastructure detection has various applications which have led to the improvement of intelligent RSUs. RSUs can

perform vehicle and pedestrian detection using information provided by cameras, radars, and the cloud in addition to V2X communication. The sensors in the RSU will have a complete view of all vehicles and other objects on the road. Thus, the RSU can handle the restricted detection capability of a vehicle by sharing its sensor data with the vehicle. This data can be utilized by the vehicle on the road for preferable trajectory planning of an autonomous vehicle, for example. The sensors mounted on the infrastructure can aid the mmWave communication system in various manners as in [89]. This paper shows that information extracted from these infrastructure sensors reduces the overhead of configuring mmWave arrays for the vehicles. To obtain accurate CSI in dynamic mmWave networks due to extreme signaling load is difficult. By sending ISAC signals, the BS sends data to the UEs and simultaneously estimates CSIs from the echoes. This approach eliminates the need for redundant pilots and the associated overhead can be effectively reduced. Reference [90] considers ISAC beamforming optimization for robust transmission in dynamic mmWave networks, where the BS transmits ISAC signals for both CSI acquisition and data transmission. At first, the ISAC beamforming design is formulated as an optimization problem aiming to minimize the worst-case sensing error subject to robust transmission rate constraints in the presence of CSI acquisition error, then a benchmarking algorithm is proposed with successive convex approximation.

A communication-centric ISAC system operating in the 60 GHz band is presented in [91]. In this paper, the environment is sensed using the information obtained from channel equalization. This work incorporates validated measurements and a customized in-situ cluster delay line mmWave channel model and shows the effectiveness of this communication-centric ISAC design. The authors study the problem of sensing task allocation in a mmWave backhaul network with ISAC in [92]. By considering target/user distribution, channel fading, and radar cross-section fluctuation, they determined the sensing duration required by each BS, if assigned sensing tasks. To handle the computational complexity in searching the solution, a sensing-oriented column formation (SOCG) scheme is proposed, which is indicated to accomplish near-optimal performance through wide performance evaluation. Furthermore, the evaluation results show that the obtained sensing task assignment ensures good throughput performance while providing that the requirements of the sensing tasks are met.

Gbps communication speeds can be achieved in the millimeter wave band using beamforming techniques with the help of antenna arrays, but a narrow beam is usually used to compensate for the high path loss in mmWave communications, and the beams at the transmitter and receiver need to be aligned to achieve high array gain. Furthermore, limited by the strong directivity of mmWave, real-time mmWave beam direction regulation is needed to improve the high reliability, low delay, and precise sensing performance. In summary, the application of mmWave beams in autonomous driving scenarios can provide high communication rates, but their strong directivity requires the design of high-performance beam tracking algorithms to achieve low-delay, high-accuracy beam switching. The authors propose a fast, robust communication and sensing mmWave beam tracking algorithm that uses the reflected echoes of ISAC mmWave beam in the case of low dynamic motion and the camera in the case of high dynamic motion as data sources in [44]. Thus, using multi-source sensing information for beam tracking, this joint algorithm and testbed increase the communication throughput and diminish the communication delay with less computational overhead for mobile vehicles. The authors focus on the mmWave massive MIMO ISAC systems that can communicate with a UE while simultaneously detecting a target in [93]. For detection and communication goals according to the proposed double-beam codebook design and beam sweeping schemes, the ISAC transmitter directs the signals towards the sensing target and communication UE concurrently, and the ISAC receiver formulates beams to receive the echoes reflected from the sensing target, effectively. To accomplish this goal, new simultaneous beam scanning schemes are proposed to find the optimum transmit and receive beams of ISAC, where both synchronous single-beam and dual-beam scanning schemes are proposed, which can accomplish different trade-offs between beamforming gain and the time overhead required for beam scanning.

The vast bandwidth available at THz (0.3 THz–10 THz) frequencies could mitigate the spectrum scarcity problem of existing wireless networks and reach terabit per second (Tbps) data rates. Moreover, the smaller antenna size allows more compact MIMO antenna designs, enabling narrower beam width with higher energy concentration in a single far-field direction. However, some key critical challenges are available in the THz frequency range. One of the most important challenges in this frequency range is undoubtedly the huge canonical free-space path loss, which significantly shortens the communication and sensing distance. In this short range, far-field planar wave assumptions may not hold. In addition, canonical beamforming in MIMO is less efficient in the near field, as beams that are narrow in the far field become wider in the near field.

Key challenges in THz ISAC and some existing algorithms to overcome them are presented in detail in [94] and [95]. ISAC in the THz frequency range is generally used for very high-speed close-range communications and for sensing smaller objects, such as human postures and gestures, and even molecular vibration. A sub-THz ISAC prototype is proposed in [96] for mobile or wearable devices with a metal lemon core imaging example presented. In this study, a THz sensing prototype with millimeter-level imaging resolution is built considering the physical aperture constraint of typical mobile device. The work in [97] proposes a system-level THz/mmWave beam alignment method that uses synchronization signal block-based detection to help beam switches. It is shown that the proposed method diminishes

the probability of beam misalignment in highway and urban situations by up to 70 percent. Therefore, detection-assisted networks that guarantee the same beam alignment quality can utilize narrower beams to expand the network coverage. [98] investigates the capabilities of THz-ISAC to simplify networked sensing for ambient intelligence applications in 6G systems. Networked sensing architecture coordinates UAVs and ground sensor networks to achieve real-time communication and environmental perception over a wide geography. Heterogeneous sensing integration methods are explored to effectively coordinate distributed sensing capabilities. A hybrid data fusion design is presented in order to integrate heterogeneous data effectively for the distributed model.

5) TRADE-OFF ANALYSES FROM A COMMUNICATION-CENTRIC ISAC PERSPECTIVE

In an ISAC paper [38], it is shown that the trade-off between communication and detection performance depends on the transmission power shared between the communication and radar missions. An optimum power allocation scheme is obtained, where eight different scenarios are considered. All of these scenarios assume that communication signals, communication channels, and radar channels are already known. In particular, [99] proposed the concept of the Communication-Sensing (C-S) Region for Cell-Free Massive MIMO ISAC systems. They provided a comprehensive analysis of the trade-offs between communication and sensing by characterizing the achievable performance region and designing advanced transmit beamforming strategies. Their work offers a holistic optimization framework that simultaneously enhances both communication sum rate and sensing performance (in MSE). This study forms an important foundation for understanding and optimizing the performance trade-offs in ISAC networks. Notably, the work by [100] proposes a unified performance metric based on the Kullback-Leibler Divergence (KLD), enabling a joint evaluation of the communication subsystem's error rate and the sensing subsystem's detection probability. The results demonstrate the inherent trade-off caused by shared resource allocation between sensing and communication, providing a comprehensive framework for optimizing both functionalities. Additionally, [101] establish a KLD-based ISAC design paradigm, unifying constellation optimization (via Deep Learning) and beamforming to quantify communication-sensing trade-offs under η_1-weighted flexibility. As another study, true-time-delay (TTD)-based 3D beamforming in [102] demonstrates how THz beam squint can be exploited to balance communication rate and sensing coverage, filling a critical gap in wideband ISAC design.

B. SENSING-CENTRIC ISAC APPROACH

In the sensing-centric ISAC scheme, radar operation is treated as a primary objective of the system, as illustrated in Fig. 7. Inspired by the wide-range applications of ISAC systems

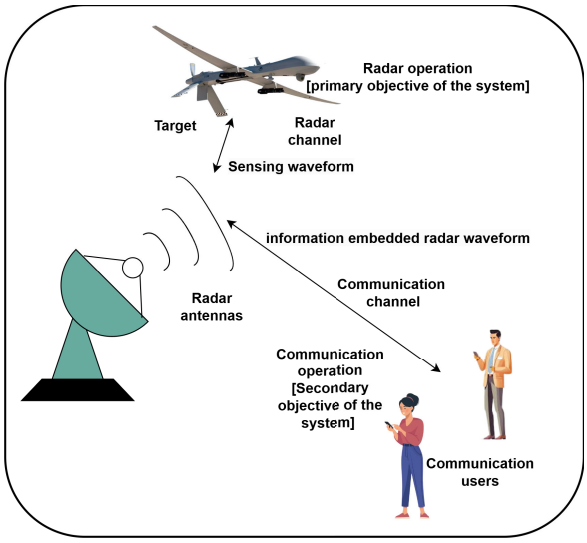


FIGURE 7. Illustration of the sensing-centric ISAC. In the sensing-centric ISAC, the antenna (or multiple antennas) functions like a radar system, detecting the target while simultaneously enabling users' communication.

across 6G wireless communication networks, we also introduce some significant survey papers here. An extensive survey on the fundamental limits of ISAC systems is provided in [35]. Firstly, the authors propose device-oriented classification methods for ISAC-related technologies and summarize the metrics, for sensing and communication performance. After current research advancements are presented, the unsolved problems are also reviewed. This subsection will discuss dual-function radar communications, the trade-off between communication and sensing, and RIS-aided reflection and beamforming in the context of the sensing-centric ISAC approach.

1) DUAL-FUNCTION RADAR-COMMUNICATIONS

Numerous works investigating dual-function systems with joint radar and communication platforms are in the literature. In [103] a dual-function radar-communication scheme is presented, using up-chirps and down-chirps as carriers. In this work, the communication data is modulated using binary phase-shift keying (BPSK) and subsequently embedded into the radar waveform through inter-pulse modulation. It is also shown that interference is relatively low due to semi-orthogonality between up and down chirp signals. In addition, the authors design an ultra-wideband (UWB) multifunctional communication and radar system. It is also reported that the studied system has been simulated, produced, and tested. The system operates on a 500 MHz bandwidth and a 750 MHz center frequency. With the simultaneous operation of communication and radar systems, the detection probability was measured as 99% with a false alarm probability of 7%. The designed UWB multifunctional system enables both monitoring of the environment and the ability to communicate.

Using a novel waveform design, the authors in [104] were motivated by multi-functional radar systems for high data rate, long-range, non-invasive communication, despite their wide bandwidth and higher power. The new coherent waveform is produced by modulating an LFM signal with minimum-shift keying (MSK). The effects of the ambiguity function (AF) and the waveform parameters on performance are examined. By keeping the total pulse width and LFM coefficient constant, theoretical analysis and simulations were presented, where it has been shown that the AF of the modulated LFM performed better.

In [45], the authors studied an ISAC system that performs both detection and communication functionality using the same hardware, thus providing more efficiency in terms of size, power, and cost. The proposed system exploits randomness in the radar scheme to transmit information to a remote receiver using index modulation (IM). In particular, the ability of the designed system to transmit digital messages is integrated into the radar system and does not cause mutual interference problems. It has been shown that the achievable bit rate of the proposed communication technique is comparable to that achieved with dedicated communication waveforms, without interfering with radar functionality. Numerical results illustrate that the proposed system exhibits excellent communication performance and effectively balances computational load and communication reliability. In [105], the authors present a radar target detection and localization scheme using space-time coding. The generalized likelihood ratio test and CRLB for target direction estimation are derived. Additionally, the localization performance is evaluated and studied both theoretically and through simulations.

A new waveform design and channel estimation solution for frequency-hopping (FH) MIMO communication is proposed in [51]. In FH-MIMO-based ISAC architectures, channel parameter estimation is challenging due to rapidly changing hopping frequencies and TMOs. In this article, channel parameters and TMOs are accurately estimated using the proposed solution. First, a new FH-MIMO radar waveform was designed to allow the receiver to detect the hopping frequency sequence (HFS) used by the radar. It has been observed that the new waveform does not cause any deterioration in radar range performance. Then, two estimators for the TMO were developed, and the mean square error (MSE) lower bound of each estimator was derived. Using these bounds, a feasible HFS is designed for both estimators. Additionally, a channel estimation method has been developed for TMO. The proposed estimation technique enables the ISAC performance to reach the approximation accomplished with perfect timing and ideal channel information, as verified by simulations.

A new technique has been developed in [106] for a dual-function system with common radar and communication platforms. Multiple orthogonal waveforms are transmitted simultaneously to insert a sequence of bits during each radar pulse. Two weight vectors are designed to obtain a spatial power distribution pattern. The proposed technique allows for information distribution to be outside of the main lobe of the radar. The waveform diversity supports MIMO configuration, where BER performance was also investigated.

Multi-antenna integrated ISAC systems are studied in the following two papers [46], [107]. The work in [46] analyzes the sensing and communication performances achieved by a MIMO downlink ISAC system. Three ISAC scenarios were investigated, including Pareto optimal, sensing-centric, and communication-centric designs. For each case, mathematical derivations for the diversity order, sensing rate (SR), and communication rate (CR) are provided for further insight. In the system model, the dual-function sensing and communication (DFSAC) BS communicates with several user terminals while simultaneously detecting targets in the surrounding environment. Outage probability is also studied for the performance of the MIMO ISAC system. A similar scheme is also reported in [107], discussing both communication and sensing performance. In this work, the DFSAC BS detects a single target while serving a communication user with a single antenna simultaneously. The paper proposes an improved beamforming design and analyzes its impact on detection and communication performance for a multi-antenna integrated ISAC architecture. New closed-form equations are derived for the above cases. The outage probability, SR, and ergodic CR are analyzed for each scenario.

OTFS has been proposed to overcome the challenges posed by doubly dispersive channels. Similar to pulsed-radars, the OTFS also operates in the delay-Doppler domain, making it an ideal candidate for the solution to ISAC problems. In [108], the authors propose a novel ISAC transmission scheme based on spatially spread MIMO-OTFS, where the angular domain is discretized by employing spatial spreading and despreading. Additionally, an emphasis is given on the effect of antenna power allocation on the performance. In [109], a triangular spectrum matching algorithm followed by a coarse search is proposed for joint target detection and range/velocity estimation, with very low computational cost. In [110], a novel technique for joint communication and sensing is introduced. In this method, the authors design an echo-preprocessing method that first removes the disruptive effects of communication symbols on the radar. This echo-processing is performed in the time-frequency domain, followed by optimizing the sensing parameter through the optimization of a key parameter.

In [52], an amplitude calibration algorithm is proposed to estimate fractional delay and Doppler parameters, which uses a barycenter formula to enhance the estimation performance of pilot-aided channel sensing. In [111], the authors improve the algorithm by employing different receive windows followed and develop a technique that explores the high-order spectra of the delay-Doppler channels. In [55], a new algorithm that formulates parameter association, channel estimation, and signal detection as a constrained bilinear

recovery problem is presented, where the problem is solved by using a message-passing algorithm. Reference [112] proposes a new method for combined radar and communication processes. This scheme transmits a single pilot signal with a rectangular pulse shape to estimate range and velocity. It then uses the detected channel parameters for channel equalization and facilitates communication. The study in [113] introduces a novel approach for ISAC in MIMO-OTFS systems by incorporating a generalized likelihood ratio test for multi-target detection.

2) TRADE-OFF BETWEEN COMMUNICATION AND SENSING

In the following, several studies on the trade-off between the sensing performance (detection, estimation, and recognition) and the achievable communication rate are reviewed. The authors in [114] focus on the communication and the sensing performance trade-off between a communication receiver and a radar receiver. The trade-off is characterized by the detection probability (P_d) and the false alarm probability (P_{fa}) boundaries against the information rate regions. A new mathematical equation for P_{fa} is derived, while P_d is approximated under high SNR assumption. The analytical formulations and the simulation results for P_d and P_{fa} are presented. The study in [47] proposes a mono-static ISAC waveform design with low complexity. The recommended design primarily considers power control. The rigorous outputs are shown while accounting for the interference issue. Specifically, for sensing functionality, the ambiguity function and P_d expression are obtained, while for communication purposes, spectrum efficiency and the symbol error rate are calculated.

A series of beamforming techniques are proposed in [115] for a MIMO joint radar-communication system, where two operational scenarios are studied. In the first scenario, one of the antennas is used for radar and the others are dedicated to communication purposes. Under this scheme, the radar signal matches with the communication beam pattern. In the second operational option, all the antennas transmit a joint waveform. The paper in [48] discusses the problem of generating a new transmission beam for an ISAC scenario consisting of a BS, a single communication user, and multiple detection targets with the eavesdroppers' imperfect CSI. The authors also solve the optimization problem related to the secrecy analysis. The transmit beam patterns and sensing beam pattern errors are shown. The authors in [34] review the ISAC systems considering the sensing and communication-centric designs. Then the enabling parameters used for design are investigated. Furthermore, they describe the tools they utilize for their design in industrial applications.

The techniques for sensing the surrounding environments and exchanging messages with the vehicular nodes for automating and coordinating in connected vehicular networks are investigated in [116]. Here, a general ISAC model is proposed with an imperfect CSI. Capacity-distortion trade-offs are also studied with numerical examples. An ISAC scheme in [49] proposes to use the CRLB model to perform the fundamental trade-off for sensing and communication functionality. The paper in [117] studies the MIMO-ISAC system. The point and extended target sensing models are inspected in this study. The maximization problem over the MIMO model is evaluated with the transmit power constraint. The authors in [118] emphasize the need for systematic performance specification criteria and appropriate assessment procedures for sensing-as-a-service in wireless communication systems. Therefore, they propose systematic sensing capability performance criteria consisting of sensing capacity, sensing sensitivity, and sensing probability. Sensing probability and sensitivity constitute the quality of service (QoS) used to express the sensing capacity. Sensing capacity reflects the overall sensing performance, specifically the number of detectable targets per unit time while considering the detection QoS requirements.

3) RIS-ASSISTED JOINT REFLECTION AND BEAMFORMING DESIGN

To enhance system throughput and spectrum efficiency, researchers have shown significant interest in using RIS to assist ISAC systems, as described in detail below. The work in [119] considers a RIS-aided ISAC system model aimed at maximizing radar SNR under the constraints of communication SINR, total transmit power, and RIS reflection coefficients. Reference [120] investigates RIS-assisted ISAC system in a cloud radio access network scenario. This work aims to generate the desired reflective beam pattern from the active RIS towards detection targets for the remote radio units. The performance plots are also provided for the reflective beam pattern MSE. RIS-aided ISAC system in mmWave communication is studied in [10] with a focus on the interference echo signal. The trade-off analysis is performed with the aid of particle swarm optimization. Another solution for the RIS-aided ISAC system is given in [121]. By taking communication QoS into account, the aim is to maximize radar SNR using RIS reflection coefficients and transmit waveform. For the non-LoS (NLoS) region, the RIS-aided ISAC solution is expressed in [122] with the goal of beam pattern gain maximization. The work in [11] investigates a RIS-enabled ISAC system. The authors propose a joint design of the transmit beamformer in the ISAC BS and the phase shift values at the RIS to maximize the user-achievable rate under imperfect angle information and CSI. Different beamforming problems with CSI error models are considered, aiming to minimize the MSE while also maximizing radar information. It is shown that the radar mutual information increases with maximum transmit power when the communication rate is kept constant.

In [12], physical layer security (PLS) is studied in a RIS-assisted ISAC system that simultaneously serves multiple users and tracks a target. The radar target is assumed to be an eavesdropper also. The goal is to maximize the achievable

secrecy rate of all legitimate users by jointly designing the transmit beamforming and the phase shift values of RIS. For this purpose, they adopt a deep reinforcement learning (DRL) algorithm. The convergence performance of the proposed optimization algorithm is presented for different transmit powers for various secrecy rates and transmit beam patterns. In [123], an active RIS-assisted ISAC system is taken into consideration which consists of an active RIS, legacy users, and a ISAC-enabled BS. The paper discusses beamforming matrix optimization in BS and reflection coefficient in RIS to maximize radar SINR, subject to QoS constraints. RIS-aided ISAC model with deep learning (DL) channel estimation problem is presented in [13]. PLS of RIS-aided ISAC systems is investigated in [124], in which the high-power radar signal is used as an interference to disturb the eavesdropper. To minimize the probability of eavesdropping, the authors formulate SINR beamforming optimization solutions for BS and RIS units. Some optimization techniques are exploited to find the proposed solution based on the fraction and semi-definite programming approaches. Numerical results show that the eavesdroppers' SINR values are efficiently suppressed thereby providing adequate QoS satisfaction levels at the legitimate users.

In [125], the authors focus on a system model featuring hybrid RIS composed of passive reflective elements and active radar sensors and numerous passive RISs to develop an ISAC system where the LoS paths between the direct BS and users/targets are obstructed. They formulate an optimization problem for the RIS phase shift matrices and transmission covariance to improve the system performance. The block coordinate descent technique is employed to separate the transmission covariance and the RIS phase shift matrices. The Riemann gradient descent approach is utilized to optimize the phase shift matrices. Another study presents two new waveform designs for RIS-enabled ISAC [126]. One of the designs focuses on spectral efficiency by maximizing communication weighted sum rate and minimizing detection beam pattern approximation error.

In brief, ISAC scenarios in the literature introduce explanations that leverage RIS's potential to shape the electromagnetic environment to advance both sensing and communication operations. In terms of enhancing detection performance, RIS can increase radar detection accuracy by focusing signals on targets. At the same time, it can support environmental detection functions with low power consumption. From the perspective of improving communication performance, RIS strengthens communication links by increasing the SNR. It is also effective in optimizing spectrum sharing for ISAC. Since sensing and communication functions in ISAC systems use the same frequency band, RIS improves spectrum efficiency by optimizing the interaction between these functions. As a powerful architecture, RIS meets the fundamental requirements of ISAC systems, namely spectrum efficiency, detection accuracy, and communication capacity. In the near future, the potential of RIS in ISAC applications is expected to be further enhanced through innovative control mechanisms, AI-based optimization, and adaptive design principles.

III. NEW APPROACHES IN ENABLING ISAC

A. JOINT WAVEFORM DESIGN AND SIGNAL PROCESSING SCHEME

Beamforming and waveform design are key research areas for ISAC systems. This subsection considers existing solutions for designing intelligent and joint waveforms using ISAC. Particularly, using a joint waveform for radar sensing and communication guarantees spectral efficiency on a single hardware platform, which can be called a single waveform ISAC-enabled BS. A single waveform can meet the requirements of both communication and radar systems [1]. An initial joint waveform was introduced for radar and communications purposes in 2006 in [127]. This paper aims to transmit the post-processed radar image as an OFDM waveform to other radars in the same frequency band.

OFDM and OTFS waveforms are widely used for designing ISAC systems. They share common properties, such as subcarrier orthogonality, wideband communication support. High bandwidth enables high-throughput communication along with high-resolution radar. The authors in [128] utilize a positioning reference signal (PRS) for sensing applications besides communication. By incorporating PRS into the 5G downlink radio frame, ISAC-BS can estimate both the target's locations and directions (angles). Different from [128], the authors of [129] claim that the transmitted data is already known in ISAC-BS and therefore pilot/reference symbols are not needed for detection. In this paper, single and multi-target scenarios are taken into consideration for a mono-static MIMO ISAC system where an OFDM waveform is employed. The transmitted signal is random due to the transmitted bits. This data-dependent effect on the received signal is first compensated. Then, the 3D constant false alarm rate (CFAR) method is applied to estimate the angle, range, and velocity values of the targets.

In [53] and [54], BER analysis is presented for a DFT-spread (DFT-s)-based ISAC system. The first paper uses the OFDM waveform, whereas the latter uses the OTFS waveform. Reference [53] proposes a DFT-s OFDM waveform for the ISAC to reduce the PAPR. The authors also propose an ISAC receiver that employs DL techniques to recover transmitted data and to estimate target range and velocity. Their simulation results show that millimeter-level accuracy in range estimation and decimeter-per-second-level accuracy in velocity estimation can be achieved. In addition, the proposed system requires approximately 5 dB less SNR to achieve a 10^{-3} BER compared to the conventional OFDM approach. A DFT-based OTFS ISAC system is proposed in [54], in which the authors compare OFDM, DFT-s OFDM, and OTFS systems in terms of BER, PAPR, and MSE of target ranges. DFT-s OTFS and DFT-s OFDM systems have almost 3 dB lower PAPR values than conventional OTFS and OFDM approaches. Moreover, OTFS provides a 1 dB lower PAPR value. The authors in [54] also investigate the change of

PAPR with the increase of the power of superimposed pilots. They observed via simulations that PAPR increases when the power of superimposed pilot signals is augmented. DFT-s OTFS provides the best performance in terms of MSE results for target distance estimation. When the target speed is above 300 km/hr, the performance of the OFDM system deteriorates and becomes saturated.

Performance constraints for ISAC waveform design, such as SINR, CRLB, and waveform similarity, are presented in [130]. Another study proposes a new approach that uses non-orthogonal waveforms, called spectrally efficient frequency division multiplexing (SEFDM), to facilitate sensing and communication tasks [131]. The system model covers multi-user MIMO-SEFDM communication and digital MIMO radar sensing, considering both omnidirectional and directional beamforming scenarios. Based on the experimental study and analysis, the trade-off and bandwidth compression factors can be adjusted to meet the communication and sensing performance requirements in specific application scenarios.

Optimization techniques are used to optimize waveform design for both communication and sensing purposes in [33] and [132]. In [33], the beam pattern for joint communication and radar sensing is optimized using a semidefinite relaxation technique. Conventional sensing beam pattern matching and the newly proposed minimum weighted beam pattern gain maximization are used for radar sensing. The JSC waveform design is investigated for correlated and uncorrelated waveforms in [132]. The trade-off between communication and sensing is also presented in that context. The authors of these papers assume that the direction of the target is known. Using this a-priori information, an optimization problem is designed to produce an optimal waveform. In most papers, a predefined user ratio and/or a power sharing between radar and communication are used as constraints for optimization problems.

Power allocation for the ISAC system is discussed in [133] under two constraints: the SINR for communications and the error performance for estimation. In this paper, the authors assume that the system model consists of multiple ISAC transceivers, sensing receivers, CUs, and a single target.

Reference [22] assumes that a BS with multiple antennas serves users over an mmWave channel and also detects targets. Since sensing the target affects communication performance, channel estimation and target detection are performed together. In [3], the authors assume that the communication channel and the sensing target angles are known to the transmit access points for beamforming design. Another study investigates both target detection and BER for uplink and downlink scenarios in ISAC using OTFS modulation [134]. In the simulation, the authors determine the locations of the RSU and four vehicles. Based on these locations and the random velocities of each vehicle, they estimate the subsequent vehicle positions and new angles of arrival. They also propose a new symbol placement scheme for OTFS to increase the data rate.

RIS-aided ISAC system performance is investigated in [119]. The optimum RIS phase shift, radar, and communication beamforming matrices are found by using various optimization techniques. The authors show that the presence of RIS in the system improves radar SNR performance. They also demonstrate that as the communication performance improves, the SNR of the radar decreases. Three scenarios are considered: No RIS case, random phase shift for RIS, and optimum RIS with a phase shift matrix. In all scenarios, the optimum beamforming matrix is calculated. In another study, a RIS-enabled ISAC system model for joint waveform design for communication and sensing is examined [126]. Two optimization problems, sum rate maximization and power minimization are considered. In the power minimization problem, the total power is allocated between communication users and sensing operation under a predetermined QoS for communication users. Simulation results demonstrate the positive impact of RIS in the sum rate analysis. Moreover, as in similar works, the sum rate for communication decreases as the number of radar antennas increases, while the total number of antennas in the ISAC BS remains constant.

B. EDGE COMPUTING

Mobile edge computing (MEC) [135], [136] was proposed to improve the computation experience of mobile users (people or machines alike), particularly in terms of total delay, energy consumption, or a combination of such metrics [137], [138]. MEC relies on ''edge servers'' installed on suitable components sufficiently close to the mobile users. These servers handle the computation of tasks generated by mobile users, who typically have limited computational power. When applied appropriately, MEC is expected to outperform both straight-forward local execution of tasks on mobile devices (by providing high-capacity servers) and cloud computing (by being available in close to mitigate high transmission delays). Some scenarios from the literature to be explored in the following subsections are depicted in Figure 8.

The interplay between ISAC-capable systems and computing is both intricate and important. In this context, the computing component could be related to ISAC procedures, such as sensing with computation-heavy schemes such as DL and DRL, or it could be unrelated applications where ISAC facilitates MEC to improve offloading performance. To emphasize this complex relationship, computation is sometimes also integrated with sensing and communications, and such systems are often referred to as ''integrated sensing, computing and communications'' (ISCAC [139], or ISC^2 [140], or SC^2 [58]). An overview of ISAC, integrated computation and communication (ISCC), and integrated sensing and computation (ISA''C'') is provided in [141], with emphasis on performance metrics and use cases.

1) MEC HELPING ISAC

Most studies treat MEC as a facilitator for ISAC. In such studies, sensing data is generated and is either processed

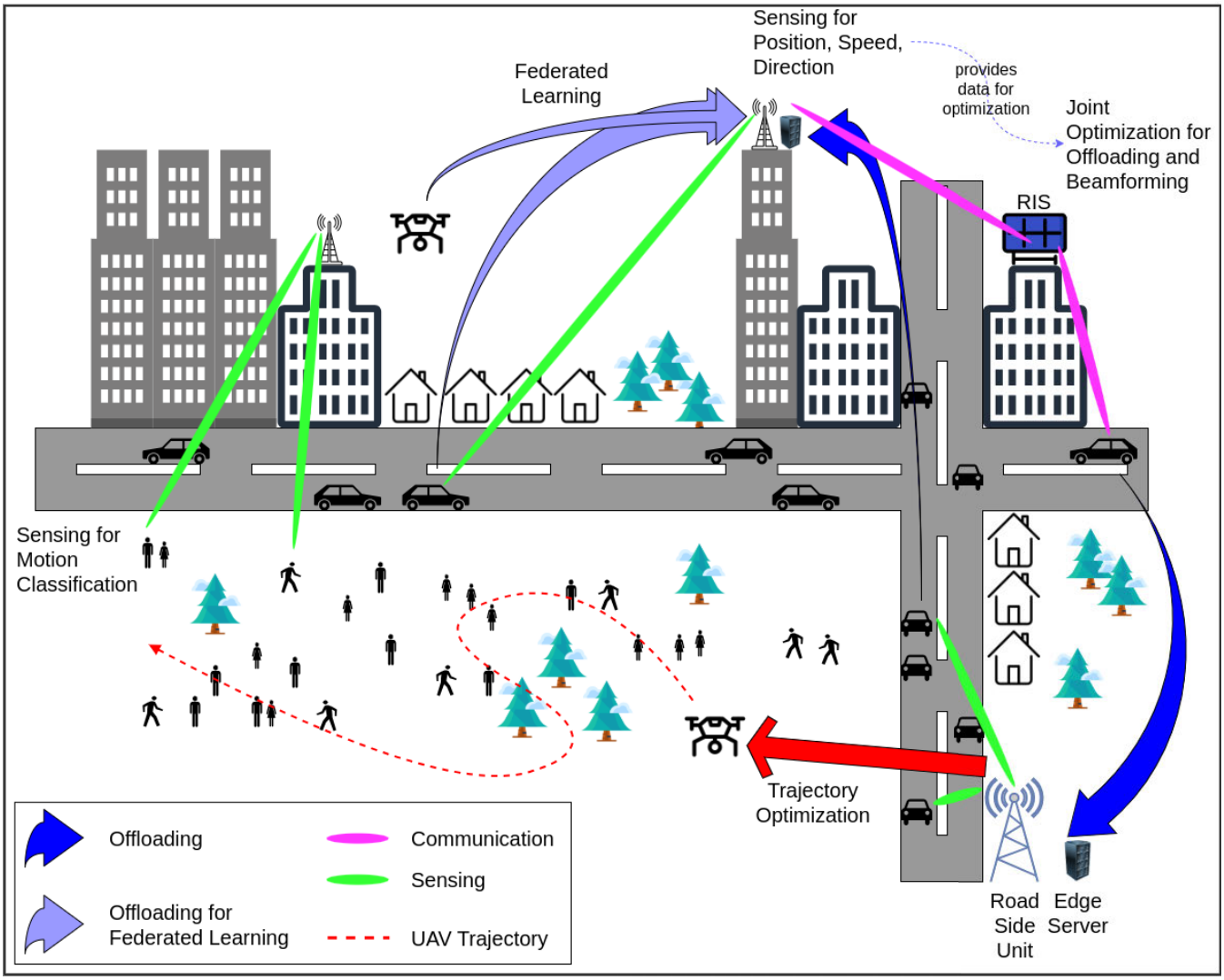


FIGURE 8. Sample scenarios with ISAC and MEC acting together and facilitating each other.

at the edge or fed to a learning system at the edge as training data. The impact and challenges of ISAC from a network-level perspective are surveyed in [142]. In this work, MEC is presented as a solution for sensing. Selecting a suitable edge server when they are sparsely deployed is revealed as a potential challenge. The work in [58] presents a federated edge learning (FEEL) scenario, where ISAC is a use case, providing the data for learning. Joint resource allocation in SC^2 is formulated to optimize the performance of FEEL. Complete sensing and communication models are defined, in contrast to most of the studies in the literature considering MEC alongside ISAC. A radar-sensing system, in which tasks are offloaded to an edge server is investigated in [143]. RIS are used to mitigate the attenuation between users and the edge server, and several parameters related to both uplink encoding and beamforming as well as MEC are jointly optimized. A federated learning-assisted ISAC system is proposed in [140] for security purposes. The main objective is to improve the security of the distributed computing system against data and model tampering attacks. A blockchain-based framework is described, which is shown to be resistant to poisoning attacks. Experimentation in this study is based on part of a measured ISCC dataset.

Mobility management in AI-driven networks with ISAC capability is studied in [144]. A layered network architecture with (i) a terminal perception layer, (ii) an edge BS processing layer, and (iii) a cloud data center layer is considered, with particular emphasis on handover mechanisms aimed at minimizing the signaling overhead. While the ISAC model is not detailed, CSI is assumed to be obtained via sensing. A simulation-based analysis is performed to demonstrate that unsupervised learning yields better spectral efficiency compared to supervised learning. Distributed edge computing is highlighted as an essential component of the architecture to meet high computation requirements, as well as a potential challenge for future AI-driven ISAC networks.

Digital twin (DT) is another feature that appears in previous works relevant to MEC and ISAC. A user-side lightweight and an edge-server large DNN are used together in a DT-based ISAC system in [145]. The lightweight DNN is sporadically updated, which is referred to as continual learning. Sensing data offloading decisions and sensing data selection for continual learning are optimized to minimize the average computation cost. DT is used to predict data drifts under different sensing data selection profiles. On the other hand, the offloading decision is very primitive, randomly made based on an offloading probability parameter.

Aerial and vehicular networks, as discussed thoroughly in Section III-C in a broader context, also constitute important use cases where we see MEC alongside ISAC systems. A review of air-ground ISAC is given in [146], where UAVs are assumed as relays both for communications and sensing. In this air-ground network, UAVs perform preprocessing in addition to sensing tasks, and terrestrial BSs process offloaded tasks for data fusion. Hotspot areas that need to adapt to dynamic traffic demands, disaster relief, battlefields, and non-infrastructure areas are listed as potential use cases. Transceiver and waveform design pattern candidates and PLS issues are briefly explored. In [147], a V2X network consisting of smart vehicles, a UAV capable of computation, and RSUs are studied. Sensing data from the vehicles are first offloaded to the UAV, and then partially offloaded to multiple RSUs for data fusion. The proposed scheme is shown to improve energy consumption for the UAV as well as task delays.

The trajectory of a UAV serving as an IoT platform is jointly optimized with communication and computation resources in [148] to minimize the weighted average of the energy consumption of the UAV and the users. A two-tiered offloading scheme is used, where the users offload sensing data to the UAV first, which further offloads to a fixed-edge server on the ground if necessary. A similar system is modeled in [149], where the UAV itself performs the sensing and offloads the data to an edge server. The trajectory, the sensing and communication power levels, and the sensing schedule are jointly optimized for sensing performance in terms of the radar estimation information rate.

An ISAC-assisted vehicular network with RSUs acting as edge servers is studied in [150]. Some tasks on particularly congested sections of the road are offloaded from heavily-loaded RSUs to lightly-loaded RSUs, where idle vehicles are used as communication relays. Delay-sensitive tasks are prioritized by timing guarantees, and overall system delay is improved. A time-division-based ISAC vehicular network is studied in [151], which also presents results from a testbed. Vehicles share sensing data among themselves and serve as edge servers to each other. The age of sensing information is also analyzed. A vehicular edge computing scenario is also investigated in [152], in which ISAC is mainly used for assisting the vehicles to sense the surrounding environment in real-time. A novel adaptive offloading scheme is proposed to cope with the large amount of sensing data generated. Reported results show improved performance in terms of both delay and energy consumption.

As MEC and ISAC have separate parameters pertinent to their separate performances, many works elected to jointly optimize both mechanisms. In [153], task latency is minimized by jointly optimizing the beamforming and the offloading processes under sensing performance constraints. This study involves a tiered computation structure, as tasks can be offloaded to a cloud server in addition to multiple edge servers. Although ISAC is not mentioned by name in [154], a system in which MIMO-capable users perform simultaneous communication and sensing via different beams for the respective waveforms using ISAC is investigated. Joint resource allocation for beamforming and energy consumption is carried out by multi-objective optimization. A multi-server MEC system is considered in [155], where sensing data is partitioned and offloaded to edge servers to be processed. Precoding matrices for communication (data offloading) and radar waveforms are jointly optimized, under the assumption of perfect channel state information, to maximize the sensing data rate achieved per energy spent. In [59], sensing data obtained via ISAC is packed into short packets whose size determines the time slot of the system, and offloaded to an edge server. The short packet size also controls the latency, and hence is one of the key parameters to be optimized, among others. Both latency and system-wide energy consumption are improved, whereas a certain sensing performance for ISAC in terms of the mutual information between the target and the ISAC device is guaranteed.

Markov decision processes and proximal policy optimization are used in [156] for the minimization of average energy consumption in an ISAC network with MEC capability. Although significant gains in energy consumption are reported, the offloading decision is based solely on an offloading probability parameter, and this is not sufficiently contrasted to ''random offloading,'' one of the benchmarks used, in terms of the selected values of the offloading probability parameter. In [157], a DRL-based offloading and resource allocation scheme is proposed. An objective function is defined as the linear combination of delay and energy consumption, and a mixed integer nonlinear program is formulated for which a double deep Q-learning network-based (DDQN) method is used to solve. Although the authors demonstrate that the proposed scheme performs well compared to all local, all offload, and alternative deep Q-learning schemes, it is evaluated in a rather small system with up to 10 users, and the effect of the weights of delay and energy consumption is not explored at all.

Systems where multiple users collaborate for a common goal also make use of MEC alongside ISAC. In [158], multiple ISAC devices perform sensing and send lightly processed sensing data to an edge server, which combines them to execute a human motion classification task. Like many other works, linear frequency up-ramp chirps are used

for sensing. Inference accuracy is shown to improve under the proposed method, compared to benchmark communication and sensing resource allocation schemes. A cooperative ISAC system where multiple ISAC devices sense multiple targets is investigated in [159]. Sensing data is offloaded to an edge server, where an ML model is trained. The overall power consumption in the system is minimized by the optimization of beamforming for both sensing and communication.

2) ISAC HELPING MEC

In contrast to the works explored in this section so far, there are also works that treat ISAC as a helper to MEC. Edge AI as a general concept is surveyed in [160], where the scenarios of centralized edge learning, federated edge learning, and edge inference are studied. The work in [161] outlines a distributed FEEL scenario, in which each user shares intermediate vectors computed from raw sensing data with the other users to perform human motion recognition while preserving privacy.

The work of Zhang et al. [162] focuses on edge intelligence systems where data set generation via sensing is an integral part of the workflow. In these kinds of scenarios, executing sensing and communication separately leads to excessive delays, and thus, ISAC is proposed to improve execution time performance. A single sensor with N antennas and M sensing targets is investigated with detailed radar sensing and communication models where interference due to ISAC is also taken into account. An optimization problem to minimize the classification error is formulated and solved in two steps. First, beamforming is optimized via semidefinite programming. Afterward, the time allocation problem for each target when the beamformers are known, which turns out to be convex, is solved. An analysis of the time gain provided by ISAC compared to sequential sensing and communications is also presented. It is shown that when generating a data sample takes longer than the uploading time of that sample, ISAC always improves performance. For the case where the uploading time of the sample takes longer than sensing, a sufficient condition for ISAC to be beneficial is derived. In [163], ISAC is used to accelerate an edge intelligence system where sensor data is fed into neural networks at the edge. The authors show that as long as the sample generation duration exceeds the sample upload duration, ISAC improves the overall delay performance of the system.

Hossain et al. [164] study the use of AI/ML-based ISAC as a facilitator for edge computing in next-generation mobile networks. A ''sensing service function'' is proposed as a two-layered network function, consisting of a global sensing service function at the core, and a local sensing service function at the radio access side, both of which are assumed to have computing capabilities. An integer linear programming problem is formulated to minimize the average latency in end-to-end edge computing with task deadlines. ISAC is used to determine parameters such as the angle (for determining the existence of a LoS) and the velocity of the user. These parameters are then lumped into a single parameter as a normalized weighted sum, which appears in the constraints of the optimization problem. The presented simulation results show that this scheme outperforms ISAC-free offloading. A noteworthy observation is that LoS-only analytics perform very close to LoS and velocity analytics together.

A DT-enabled internet of vehicles system is investigated in [165]. An ''environment aware offloading mechanism'' to solve the joint optimization problem of task scheduling and resource allocation for vehicular MEC is proposed. The proposed method is based on a Markov decision process and deep deterministic policy gradient-based Q-learning, and aims to minimize the maximum overall response time of the generated tasks. In this study, ISAC is used to facilitate the offloading decisions for each task. RSU and certain vehicles in the system called service vehicles are assumed to be equipped with sensors and capable of retrieving environmental information such as the speed and the location of the vehicles via ISAC. Along with these, information on the task characteristics and available resources constitute the DTs of the RSUs and the vehicles. The main function of ISAC is to enable MEC in this study, and it is assumed that the necessary information is acquired from ISAC perfectly, and details on how this can be achieved seem to be left out of the scope of the study.

Some works in this category also use joint optimization techniques. RIS is used in the work of Huang et al. [166] where energy consumption is minimized under radar rate as well as delay constraints in an ISAC-assisted MEC system. On the other hand, some others see ISAC just as a black box, such as [167], in which a DRL-based formulation is laid out for the optimization of a joint objective function involving task delay and energy consumption in an internet of vehicles setting. This study incorporates partial offloading but has two main shortcomings. The sensing scheme of ISAC is not incorporated into the model, and the solution to the optimization problem is not provided. ISAC serves merely as a data source for a vehicular network that uses MEC to provide environmental information to the vehicles in [17]. Energy efficiency is the key objective achieved by the joint optimization of computing and communication resources.

3) ISAC AND MEC HELPING EACH OTHER

There are also a handful of works that study systems where ISAC and MEC are used symbiotically. The architecture and resource management of a 6G ISCAC network are investigated in [139]. A multidimensional layering of a network architecture with the aid of DTs is proposed. Two novel metrics, ''network service capability'' and ''economic indicator'' are proposed and described, but not mathematically defined. Reducing the age of information regarding the freshness of data used for sensing is the main focus, which is shown to be achieved via a simulation study.

Image recognition application in edge networks for ISAC systems is studied in [168]. The image captured by a wireless sensor is transmitted to an edge server for analysis in parallel with radar detection functionality. The authors define the main challenge of the system as the interference between image data transmission and radar detection, negatively affecting the performance of each other. To mitigate this problem, RIS usage is proposed. Extensive numerical results are provided to demonstrate superior performance of RIS usage with phase optimization, in terms of a general metric called sensing utility. Edge computing is treated as a service in itself in [169]. Time scheduling, resource allocation, and offloading decisions for tasks are jointly optimized for a composite objective function involving task delays and transmission rates. The processing of sensing tasks and other computation tasks are treated separately in [170] in a V2X network. A joint optimization problem for sensing accuracy and task latencies is formulated concerning resource allocation and solved using DRL. This work uses a deterministic all-offloading policy.

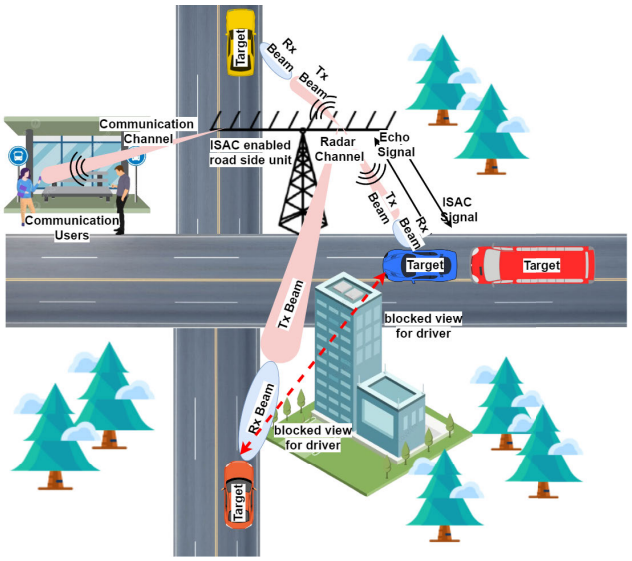


FIGURE 9. Illustration of an ISAC in a vehicular communication network.

4) CHALLENGES IN ISAC AND MEC INTERPLAY

ISAC and MEC are two key technologies expected to make considerable impact on actual 6G implementations and deployments in the near future. MEC appeared in 3GPP standards first in Release 15, whereas ISAC was included first in Release 16. Although they were conceived as solutions to separate problems, there is a significant overlap between the use-cases they feature in. MEC is a key enabler for edge intelligence and AI-native communication systems, whereas ISAC can provide the data necessary for edge intelligence.

There are many applications where MEC and ISAC are used together, in which one or the other may be more prominent and central to the application. One of the foremost issues in such scenarios is assuming a black box for the component that is not central, such as assuming that MEC handles all the computation perfectly in an ISAC system, or assuming that ISAC produces perfect data to feed into a MEC system as in [167], without paying any attention to the challenges in these black box components. This approach may also lead to simplistic solutions or assumptions such as making offloading decisions based on a fixed offloading probability parameter as in [156]. There are also works that jointly optimize aspects of MEC and ISAC, including [156], as well as [166], where energy consumption is minimized under constraints pertinent to both ISAC and MEC parameters. The main issue with these approaches lies in the fact that such studies formulate optimization problems for a snapshot of the system rather than a realistic time dependent behavior. Many studies assume a known profile of MEC tasks, usually one task generated per involved device, and ignore stochastic task arrivals. Consequently, the reported gains are valid only for a static and deterministic system.

Future efforts into systems involving MEC and ISAC together should avoid both treating one of these components as a black box, as well as assuming static computation task profiles. Handling these two key technologies will pave the way to more efficient handovers and resource allocation algorithms, in addition to enabling edge intelligence.

C. ISAC IN VEHICULAR COMMUNICATIONS

In recent years, the development of vehicular networks has evolved significantly, particularly with the advent of 5G technology building upon the 4G infrastructure and the long-term evolution (LTE) and 3GPP standards [171]. Fig. 9 shows ISAC in vehicular communications in a conceptual transmission context. This figure considers an ISAC vehicular network in which the ISAC-enabled RSU provides communication services to multiple users while sensing numerous targets in the same environment.

ISAC in vehicular networks articles first appear in the literature focusing on mmWave beam alignment and radar-aided communication. The use of dedicated short-range communication or automotive sensors is suggested as side information to reduce the mmWave beam alignment overhead in [172]. This paper displays that mmWave communication is the only reasonable path for high-bandwidth linked vehicles. The suggested idea in this article can be accomplished by attracting side information captured from various out-of-band sensors, which will show a significant role not only for mmWave V2X communications but also in achieving tolerable channel quality with shortened control overhead. Radar-aided vehicular communication for the mmWave vehicle-to-infrastructure (V2I) link is discussed in detail in [173]. In this paper, side information in a specific mmWave band is used to fit the beams of a vehicular communication scheme performing in another mmWave band which is expected to achieve the beam alignment task based on the prescribed methods.

Later, articles were initiated to show up under the headline of ISAC. A typical ISAC design for vehicular networks is stated in a tutorial on joint radar and communication transmission for three different parts [24], [25], [26]. In the first part, the authors briefly propose the background and the impending utilization of the ISAC technique, along with introducing both radar and communication architecture, covering the system design, and fundamental achievement metrics [24]. The second part includes a state-of-the-art ISAC-based beam tracking and prediction technique [25]. Next, the researchers in this tutorial series offer conventional beam training and tracking techniques in the generic cellular network. Following that, they overviewed an ISAC-based predictive beamforming design conceived for the V2I links and showed how vehicular communication performance benefits from radar sensing. In the last part of this study, the predictive beamforming for the V2I links is inspected. With the aid of the ISAC signals granted by the RSU, the beam tracking is accomplished.

The location parameters for the vehicles are approximated by exploiting the reflected echo signals and the achievable rates are provided over the designed architecture for the single and multi-antenna embedded vehicles [26]. The authors in [56] explained a radar-assisted predictive beamforming design for V2I communication by exploiting the ISAC technique. The suggested solution aims to trail the angular change of the vehicle, using an extended Kalman filtering framework utilizing the echo signal statistics as well as the vehicle's state. To minimize the tracking errors while assuring the downlink QoS, an optimization-based power allocation architecture recognizes both angle and distance estimation under the threshold sum rate. The information related to the ISAC-based beamforming scheme for vehicular networks requiring only a very low signaling overhead for beam tracking is given in [57]. To lessen the latency as well as the overhead for signaling, the probabilistic approach, expressed in this article, has succeeded in tracking and predicting the motion of vehicles.

A vehicular ISAC scenario that uses MIMO antennas is proposed in [60]. For vehicular networks, various types of services are prescribed for distinct vehicles accomplished with a study of the physical-layer service integration approach. In this solution, the ISAC BS provides public and confidential services together for vehicles and implements beam tracking based on the echo signals at a high-security level while boosting the SNR for the echo signal. The channel pattern of a multi-antenna heterogeneous vehicular ISAC system is investigated in [174], where a dual-functional BS interacts with the multi-antenna vehicular receiver and senses the enclosing surroundings together. For both mono-static and bi-static sensing schemes, the proposed ISAC channel model can be fitted. The outcomes depict that the communication and sensing channels present different propagation properties, and the sensing channel is more sensitive to mobile targets. A framework for power and bandwidth designation over ISAC networks in vehicular systems is proposed in [175]. The necessities for low-latency services in vehicular networks are accomplished for the scheduled architecture with the aid of RSUs. Reference [176] propose a method that combines ISAC and AI for path loss prediction to improve the performance of wireless data transmissions in vehicular networks.

ISAC-based path loss forecasting is designed to enhance the intelligence of wireless transmissions in vehicular networks, in which data is collected by mmWave radars, laser radars, and cameras. With the aid of supervised learning, the proposed models' scalability and accuracy are verified over the weather conditions, geographical conditions, and traffic overheads. Reference [177] presents a data symbol reservation algorithm based on a vehicular network transmission scheme using ISAC signals and OTFS modulation. In the proposed algorithm, improvement of channel estimation success and system efficiency is achieved. Reference [178] demonstrates the reduction of resource elements and scanning beam frames used in the vehicle detection stage of ISAC within a New Radio (NR) OFDM scheme. A solution to sensing-aided uplink transmission in an ISAC vehicular network using OTFS modulation is given in [55]. In this article, the authors focus on the sensing parameters on the joint parameter corporation process captured by the RSU through the downlink stage for channel estimation and signal detection.

Vehicular ISAC-assisted pedestrian recognition to improve the effective throughput is studied in [179]. The authors consider the scenario where the recognition object is the same as the communication one. Due to the sensing efficiency impact on effective system throughput, they develop a joint allocation plan of sensing duration and transmission duration for ISAC vehicular networks. Finally, a method to enable ISAC for vehicular networks by extending the 5G V2X Mode 2 protocol is reported by the authors in [180]. The focus of this study is to simplify both periodic communication data transmission and radar detection. The proposed solution can accommodate each vehicle's radar scanning period and transmit power, based on the acceleration of the vehicle and the information for channel congestion, ensuring vehicle safety.

An energy-efficient computation offloading strategy is discussed in [147] for multiple sensor data fusion in a UAV-aided V2X network supported by ISAC. UAV deployment and precoder optimization for multicasting and target detection are considered in this study. The authors inspect the affairs based on content delivery and target sensing over UAV-aided ISAC networks. To store the user-desired content, deliver it directly to the end-users, and perform target detection are practiced by UAVs. The researchers evaluated the achievement of content sending and target sensing with the help of optimization techniques. Finally, the success of the suggested user cooperation, UAV deployment, and precoding design algorithm for UAV-aided ISAC networks are displayed.

Recently, there have been studies in the literature on beamforming and radar sensing in vehicular networks. Reference [181] informs a DL-based beamforming design

for joint multi-user communications and radar sensing in V2I systems. The researchers focused to diminish the CRLB of radar sensing while assuring a tolerable level of attainable communication rate. This task concluded with the aid of neural network approaches. The paper in [182] a vehicular behavior-aware (VBA)-ISAC beamforming design for vehicle-mounted transmitters is studied. In addition, the optimization problem is formed to boost the spectral efficiency of the communication rate while degrading the mismatch for the beam pattern, in which a trade-off is realized to balance the sensing and communication performance. A DL-based beamforming design for joint multi-user communications and radar sensing with vehicles exhibiting complex behaviors in ISAC-based V2I systems is expressed in the paper of [183].

Although existing studies on ISAC for vehicular networks have made valuable progress in various domains, including beam alignment, predictive beamforming, sensing-assisted communication, and AI-supported optimization techniques, most of these works focus on solving specific technical problems within limited and isolated scenarios. It is observed that a comprehensive perspective that jointly addresses communication reliability, sensing accuracy, and resource efficiency in realistic vehicular environments remains largely unexplored. In particular, there is still a considerable need for developing ISAC frameworks capable of dynamically adapting to diverse vehicular conditions, varying mobility patterns, and environmental uncertainties. Therefore, future research is expected to focus on designing resource management strategies, practical hardware implementations, and advanced AI-based algorithms to enable reliable and intelligent communication and sensing functionalities in next-generation ISAC-enabled vehicular networks.

D. PRACTICAL IMPLEMENTATIONS AND SCALABILITY OF ISAC

Practical implementations of ISAC include high-precision positioning, localization, and tracking for applications such as biomedical imaging, security applications, complex terrain mapping, indoor and outdoor positioning and navigation, pollution and natural disaster monitoring, gesture recognition, remote sensing for agricultural automation, livestock tracking, weather sensing and forecasting, V2X integration with autonomous driving, stealth military applications and electronic warfare, and various other advanced services. These capabilities are expected to enable a wide range of applications across multiple sectors, offering benefits to future consumers and a variety of vertical industries.

While the majority of ISAC studies in the literature are theoretical and conceptual, there are also a few practical applications [184], [185], [186], [187]. In 2017, an experimental ISAC system was implemented through a software-defined radar testbed with four transmit channels [184]. The center frequency and bandwidth were set to 3.5 GHz and 10 MHz, respectively, with the assumption that the communication is only performed half-duplex. Two separate beams were produced for radar and communication purposes. The power of the communication beam was 13 dB lower than the power of the radar signal. QPSK modulation was used for communication, where an LFM signal was employed for radar. Beam pattern measurements, radar waveform validation, and BER analysis of the communication beam were presented. In 2019, an ISAC experiment was implemented as presented in [185]. The authors in this paper successfully achieved high-resolution synthetic aperture radar (SAR) imaging while simultaneously performing communication. However, they emphasized a slight interference between the communication and radar signals.

In [186], real 5G systems are enhanced with ISAC, where the sensing signals are embedded in the frame structures. Therefore, interference between radar and communication signals was addressed. The system can detect, track, and localize multiple targets simultaneously. Recently, USRP 2953R-based practical implementation has been carried out at Warsaw University in Poland [187]. The TX and RX-target separations are set to be 180 and 110 meters and the total bi-static range is assumed to be 240 meters. Both target locations and velocity are estimated successfully.

The scalability of ISAC systems for real-world scenarios can be divided into several factors, depending on their practical implementations and performance. One of the main challenges to the scalability of ISAC systems is overcoming hardware and infrastructural constraints. Scaling ISAC to dense environments, such as urban areas, requires advanced computing hardware to support high-resolution sensing and high-throughput communication. Moreover, advanced antenna and RF chain deployments in the PHY, such as MIMO and massive MIMO systems, further increase the complexity in terms of cost and design.

Spectrum management and interference mitigation are other significant challenges in enabling future ISAC systems. The reuse of the spectrum can be crucial as both high-resolution radar and high-throughput communication require extended bandwidth. With many ISAC-enabled UEs, including IoT devices, mitigating interference will become increasingly complex.

Synchronization between different ISAC network nodes is another scalability challenge. As the volume of sensing data increases, the demand for scalable backhaul also grows. Some of the most important key enablers for the scalability of ISAC can be itemized as follows.

- Joint waveform design to enable efficient spectrum usage across a very high number of devices,
- AI/ML for resource allocation and optimization,
- Edge computing for local and low latency processing, especially for distributed ISAC systems,
- RIS for scalable coverage and sensing,
- Efficient synchronization and coordination methods for coherent multi-node sensing and communication,
- Scalable beamforming using hybrid analog-digital MIMO architectures to support multi-user ISAC with

spatial reuse while minimizing design cost and energy consumption,
- ISAC-capable application-specific hardware and software that integrates both sensing and communication, which can also enable IoT-scale devices with tight energy and size constraints,

which are covered throughout this survey.

IV. 3GPP STANDARDIZATION ON ISAC

Within the context of the 3GPP standards, ISAC is a fast-expanding and promising field in wireless communications. By integrating sensing capabilities into existing communication networks, 3GPP aims to transform the infrastructure into a dual-purpose tool for both communication and environmental sensing. Additionally, ISAC standardization efforts for ISAC within the 3GPP organization are still in progress to integrate sensing capabilities into both current and upcoming wireless communication standards [19], [20], [21]. Within the 3GPP framework, ISAC is still a relatively new concept, and its integration into different 3GPP versions is an extended work in progress. The agreed-upon items, which include various scenarios, are typically use cases that are often designated. The forthcoming deliverance is foreseen to bring sophisticated and standardized ISAC approaches, expanding the range of applications and enhancing connectivity. The precise explanations of how ISAC is incorporated into various releases are provided in [23].

The 3GPP Release 16, the first to feature ISAC as a concept, was triggered in 2018 and ended in 2020. The introductory ideas about adding sensing capabilities to communication systems were enclosed in this final release. Next, a preliminary investigation of use cases is carried out, including advanced positioning, location-based services, and V2X communications. Lastly, the initial release of 3GPP for ISAC has laid the groundwork for future development on ISAC. The successive phase, namely 3GPP Release 17, commenced in 2020 and was completed in the middle of 2022. The main focus of this release is on 5G upgrades, in conjunction with new services and use cases. The technical reports and feasibility studies on spectrum allocation and integration of ISAC provide the fundamental framework for this phase. Another priority area for 3GPP Release 17 is the initial technical development of combining basic sensing capabilities with communication functionality. In summary, Releases 16 and 17 have marked the beginning of ISAC concepts' exploration and the feasibility studies for understanding the prospective advantages and technical difficulties. 2024 is the anticipated completion date for 3GPP Release 18. Resource allocation, waveform design, and signal processing for ISAC are the primary contexts of this release, where the early V2X and UAV-based technical standards and requirements will be determined. The programmed timeline for 3GPP Release 19 is 2024-2026. Incorporating the standardization of ISAC features into the official 3GPP specifications is expected to occur in this version. Besides, technical work on resource management strategies including protocols and interfaces will be conducted. Other major goals of this version include field trials and real-world testing to validate ISAC performance.

Finally, 3GPP Release 20 is expected to be finalized around 2027. Soon, this release will focus on next-generation mobile networks. Full integration of ISAC capabilities as a core component of 6G networks is planned to be live in this release. Besides, highly sophisticated sensing and communication functionalities are going to be implemented along with use-cases of ultra-high accuracy localization and tracking, terahertz communications, full integration of AI/ML approaches, extreme URLLC, massive IoT, localization, simultaneous imaging, mapping, activity and gesture recognition, and augmented human sensing. Consequently, this update aims to provide a seamless experience by fully integrating sensing and communication. By following the timeline in Fig. 10, 3GPP seeks to gradually introduce, refine, and fully integrate ISAC capabilities into their standards, ensuring that future wireless networks can provide both advanced communication and sensing functionalities including the following potential main applications: autonomous vehicles, smart cities, UAV services, health monitoring, industrial applications, road traffic applications, environmental applications, intrusion detection applications [22], [23].

V. SECURITY IN ISAC

ISAC technology is envisioned to play a key role in the upcoming generations of communications by enabling two indispensable functionalities to operate jointly in a spectrally and energy-efficient manner. However, the ISAC paradigm leads to new challenges in terms of data security and privacy due to its integrated architectural design, where sensing receivers may undesirably be exposed to communication users' data, or vice versa. This brings a raised possibility of security breach by unauthorized terminals or eavesdroppers. Such a scenario is depicted in Fig. 1, where a transmitter desires to form a secure communication link with a legitimate receiver while sensing a target (Eve) which may function as an eavesdropper. Several security and privacy challenges in ISAC can be itemized as given in the following list.

- Signal interception: Both communication and sensing signals are susceptible to unwarranted interception putting system confidentiality into risk.
- Jamming: Possible jamming attacks on the ISAC signal can degrade system performance and cause to disrupt sensing and communication.
- Data manipulation: Malicious nodes who collect data in an unauthorized way can apply signal manipulation to corrupt the accuracy of both communication and sensing functions.
- Location privacy: Due to the involved sensing process, some user/device locations can be exposed unintentionally in ISAC.
- Unauthorized access: If the suitable measures are not taken, unauthorized nodes may access the sensitive data.

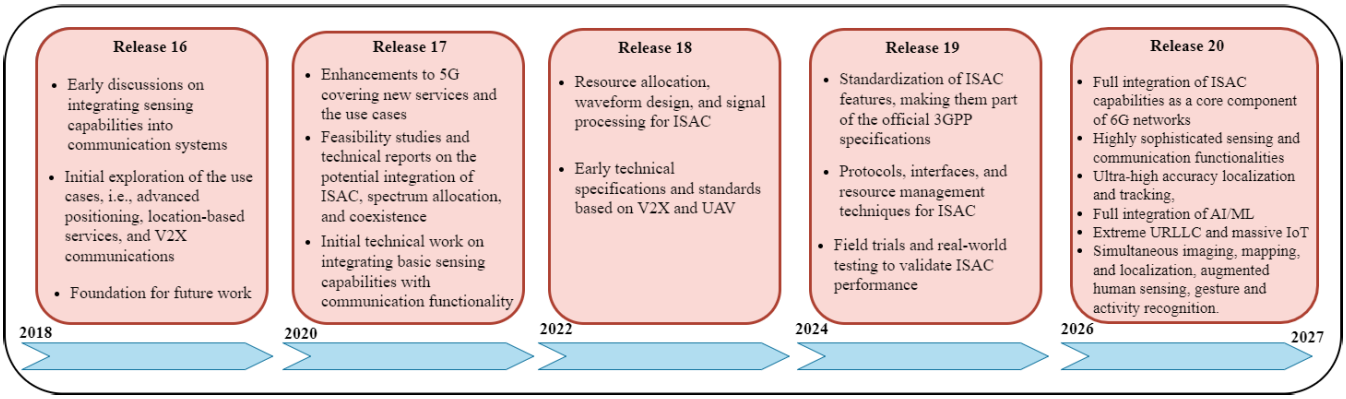


FIGURE 10. Overview of 3GPP standardization progress on ISAC, highlighting key milestones, current focus areas, and future expectations.

- Malicious tracking: Illegitimate entities may exploit available sensing capabilities to viciously surveillance and track the legitimate nodes.

In this section, a thorough survey of the related studies on the integration of security into various ISAC scenarios is provided.

The authors in [60] consider a vehicular communication scenario where a MIMO ISAC BS sends confidential messages to a certain set of vehicles with higher security levels, while broadcast messages are transmitted to all the vehicles. The transmitter also performs beam tracking by relying on echo signals reflected from these vehicles. To decrease the SINR level corresponding to the confidential messages at the lower-security level receivers, a certain amount of artificial noise (AN) is added to the transmitted signal in a controlled manner. Simulation results verify that the introduced scheme has the potential to enhance the echo signal SNR while meeting certain performance requirements. A secure multi-user downlink ISAC scenario is studied in [61] and [62] where a multi-antenna BS communicates with communication users via beamforming, while sensing certain targets which also act as potential eavesdroppers. To ensure transmission secrecy, the transmitted ISAC signal is composed of confidential information for legitimate users and a certain amount of AN. In [61], it is shown that the performance of the introduced robust technique with partial knowledge of the eavesdroppers' CSI values approaches the performance under a perfect CSI scenario. The authors focus on the resource allocation problem in [62]. Unlike previous related works, a novel optimization architecture is suggested utilizing joint optimization of communication and sensing resources over a sequence of snapshots with alterable durations. Simulation results reveal that the presented approach accomplishes better PLS levels compared to three relevant baseline methods.

Several objectives and problems encountered when endeavoring to secure ISAC systems are listed in [188]. In addition, a security framework is introduced where the knowledge acquired about the target via the sensing process is proposed as an empowering opportunity against the known vulnerabilities of PLS. In [189], the authors treat the problem of enhancing the spectrum utilization in an ISAC ultra-dense network scenario with cloud RAN configuration supporting centralized processing and capable of beam sweeping. Taking a non-cryptography perspective, a space-time decoupling technique and a challenge-response verification method for spoofing violations are proposed and are shown to be effective through numerical results. In the sensing process based on CSI, pilot signals can be acquired by a malicious user who can then obtain critical knowledge on a certain target by performing sensing. In [190], this security problem is addressed by introducing a secure pilot allocation approach to hide the pilot signal information from malicious ends. Simulation results are used to demonstrate the strength of the proposed scheme in terms of providing secure communications and sensing. The work in [191] inspects a secure ISAC system in the form of a dual-functional radar-communication scheme. Accordingly, a wireless surveillance system with radar functionality is developed, where the legitimate surveillance and radar sensing processes use the same spectrum and hardware. By integrating proactive eavesdropping and radar sensing, the transmit and receive beamforming vectors are jointly optimized. The efficiency of the presented scheme is numerically shown, concerning eavesdropping success probability.

A secure multi-user uplink communication scheme is studied in [192] under an ISAC scenario. Here, the BS sends radar signals to track and jam a mobile aerial eavesdropper. The authors resort to an extended Kalman filter method for trajectory tracking and present an optimization problem for enhancing the secrecy performance. It is demonstrated that the suggested idea attains superior secrecy capacity and fairness as compared to other contemporary related methods. In [48] and [193], the authors investigate a similar scenario as in [61] and [62] by assuming a single communication user. In addition to the information data for the communication user, BS transmits a devoted sensing signal, which also serves as AN to perturb the received signals at the eavesdroppers. Numerical findings indicate that various trade-offs can be achieved between the communications and

sensing performances while guaranteeing the demand for the secrecy rate at the communication receiver.

In [194] and [195], the secure transmit beamforming design problem is addressed for ISAC systems by considering both the existence and nonexistence of eavesdroppers' CSI. It is aimed to minimize the maximum eavesdropping SINR by satisfying certain QoS constraints such as the communications and radar detection performances. Numerical results illustrate the benefits of the proposed solution in terms of transmission security. A multi-user single-target ISAC downlink scenario is inspected in [196] to enhance PLS. To that end, the constructive interference idea, which is a symbol-level precoding approach relying on multi-user interference, is exploited. Numerical findings affirm that the proposed scheme attains an excellent trade-off among sensing, communications, and PLS performances. In [197], the authors consider a secure downlink ISAC scheme where a BS communicates with a receiver while sensing the position of a target which can behave as an eavesdropper employing its reflected echo signals. By exploiting the target's position distribution, an AN-based transmit beamforming framework is developed and shown to maximize the worst-case secrecy rate. The presented method is numerically demonstrated to perform better than miscellaneous benchmark approaches in terms of secrecy and sensing performances.

A secure ISAC scenario is studied in [198] for perceptive mobile networks, which can enable ubiquitous sensing capabilities and robust wireless connectivity with minor adjustments of the infrastructures and protocols in the present cellular systems. Taking the randomness of the transmit waveform into account, the expectational minimum mean square error is embraced as the sensing performance criteria and is optimized by considering the constraints in transmit power and secrecy rate. The study in [199] concentrates on PLS for an ISAC scenario, in which there are multiple downlink communication users and numerous targets (potential eavesdroppers) to be sensed. To ensure confidential communications, a new design of a transmit beamforming scheme is suggested regarding two classes of receivers. Numerical results confirm the effectiveness of the proposed scheme in terms of ensuring secure communications. In [200], an AN-based beamformer design problem is treated to ensure security for a MIMO multi-user downlink ISAC system which encompasses a BS, multiple legitimate communication users, various eavesdroppers, and a target to be sensed. The resulting non-convex optimization problem is slightly customized and subsequently solved by semidefinite relaxation.

A jammer-endorsed covert ISAC system is inspected in [201] using joint beamforming and power allocation under certain security and sensing capability constraints. Several numerical results are used to show the advantages of the proposed scheme as well as the effect of jamming power and the impact of sensing constraint on the covert rate. In [202], the authors investigate a multi-user ISAC system with a single BS and a unique target. To ensure secure communications and target detection, a precoding matrix and AN design problem are studied to maximize the attainable transmit rate under the beam pattern similarity constraint, the eavesdropper SINR restriction, and the power constraint. The increase in the secrecy sum rate is verified via simulation results. The work in [203] considers a state-dependent wiretap channel over an ISAC scenario where CSI is assumed to be available at the transmit side. The legitimate terminal tries to decode its message signal as well as to reconstruct the channel state. An encoding process is devised to mask the message and the channel state from an eavesdropper. The impact of the distinct security constraints on the achievable rate-distortion region is also illustrated.

A sensing-assisted PLS system is proposed in [204], [205], and [206] under an ISAC scenario where a BS sends confidential information to communication users while attempting to sense channels of potential malicious eavesdroppers. The secrecy rate expression is described as a function of the estimation precision of the eavesdroppers' channels. To maximize the secrecy rate and minimize the mean square error of estimates, the authors also work out a weighted optimization problem. In [207], a PLS optimization model is built for an ISAC system by considering the possibly imperfect CSI of the eavesdropper. CRLB is adopted as the sensing metric, and the trade-off between the sensing and communication performances is derived under PLS improvement. A secure ISAC system is investigated in [208] where a multi-antenna BS communicates with various terminals and tries to sense certain targets, which may behave as eavesdroppers. Bayesian CRLB on the sensing accuracy is derived and is used as the optimization criteria to be minimized while guaranteeing definite communications QoS constraints by utilizing constructive interference. The received signals at the illegitimate receivers are desired to fall into the destructive region of the signal constellation such that the decoding probability for the communications data at the eavesdroppers significantly decays. Simulation results confirm the success of the introduced secure ISAC framework.

In [209], a covert communication system is investigated over an ISAC scenario composed of a radar with multiple antennas, a multi-antenna warden, and a single-antenna communication receiver. The aim is to enable the radar to sense a target while communicating with the legitimate receiver covertly under the warden's surveillance. The problem of beamforming optimization is treated for the dual-functional waveform to maximize the covert rate under the constraints of covertness, transmit power, and radar detection. Simulation results indicate that the presented scheme can accomplish covert communications as well as establish a certain degree of sensing performance.

The security problem in a full-duplex ISAC system is studied in [210]. Here, a BS communicates with both uplink and downlink users and carries out a target-tracking

operation. The target acting as the potential eavesdropper attempts to intercept the information transfer in both ways. To ensure full-duplex secure communications and target sensing, the authors conceive the information beamformer, the radar waveform, the uplink communication receive filter, and the radar receive filter. Numerical findings show that the presented technique performs better than other related approaches in terms of security. An analogous scenario is also covered in [211] where a design problem is introduced to support secure communications over both downlink and uplink secrecy rate demands and illuminate radar beams towards eavesdroppers. The superiority of the presented method is demonstrated via simulation results. The work in [212] establishes information-theoretic limits for a low-latency secure ISAC system comprised of a transmitter, a legitimate receiver, and a target/eavesdropper. The rate region is determined under a degraded wiretap channel assumption. The finite block-length scenario is also considered to derive attainable non-asymptotic performance limits.

A. SECURE ISAC SYSTEMS: RIS APPLICATIONS

In ISAC systems, RISs, also known as intelligent reflecting surfaces (IRSs), enable a feasible solution to improve both sensing and communication performances at the same time through reconfiguring the radio propagation in the form of reflecting the incident signals with suitably set phase and/or amplitude values. RIS technology even allows for the formation of virtual LoS links between a transmitter and a receiver/target located in a non-LoS area. In recent years, increasing number of studies have investigated ISAC systems based on RIS from a security perspective. A comprehensive survey of such works is given below.

PLS is studied for a multi-user multiple-input single-output (MISO) ISAC system with a malicious UAV trying to eavesdrop on the communications link in [63]. To maximize the attainable secrecy rate, the use of an active RIS is advocated through a joint design of the radar receive beamformers, the active RIS reflection coefficients matrix, and the transmit beamformers at the dual-functional BS. The ensuing optimization problem is shown to be non-convex and is solved by using fractional programming as well as majorization-minimization methods. Simulation results exhibit the excellence of the presented scheme as compared to other related techniques with passive RIS and with no RIS at all. In [124], a RIS-aided ISAC system is considered in terms of PLS where a BS with multiple antennas communicates towards multiple single-antenna downlink users with the assistance of a RIS and tracks a target at the same time. In addition, an eavesdropper endeavors to listen to the transmission of confidential information. The authors provide a joint beamforming optimization framework for active beamforming at BS and passive beamforming at RIS to minimize the maximum eavesdropping SINR. Numerical findings support that the presented approach can significantly decrease the SINR at the eavesdropper.

A quite similar scenario is also treated in [12] where a transmitter conveys information to a set of legitimate users while detecting a target that tries to eavesdrop on the communication links. RIS is used to enhance the signal strength of the receivers within the coverage area. To maximize the attainable secrecy rate of all the legitimate users, the transmit beamforming, the AN signals, and the phase shifts of RIS are collectively optimized utilizing DRL methods. The numerical results verify the effectiveness of the presented scheme in terms of secrecy rate as compared to other relevant benchmark techniques. The authors of [213] inspect the problem of transmit and reflective beamforming design over a RIS-empowered secure ISAC system. The objective is to maximize the beam pattern gain in the direction of an eavesdropping target under certain QoS and privacy constraints. To deal with the resulting non-convex optimization problem, a novel second-order cone programming method based on successive convex approximation is recommended. The superior performance of the proposed scheme against a benchmark approach is numerically demonstrated in terms of the beam pattern gain.

Conventional RIS architectures can only provide coverage in certain directions. To provide full-angle coverage and to increase the available degrees of freedom, simultaneously transmitting and reflecting (STAR) RIS can be employed. A STAR-RIS arrangement is adopted in secure ISAC networks in [214] and [215]. A BS transmits a multicast communication signal together with an allocated sensing signal to accomplish multi-user communication as well as a single target detection under the existence of an eavesdropper in [214]. The authors try to optimize beamforming processes at the BS and the RIS from the perspective of maximizing the secrecy rate while satisfying certain sensing performance criteria. Numerical results corroborate the effectiveness of the presented approach. In [215], a similar case is covered where a BS carries out sensing of a target while transmitting information to a legitimate terminal in the presence of an eavesdropper. The average long-term security rate at the legitimate user is maximized by a joint configuration of the transmit beamforming at BS, the transmitting and reflecting coefficients of the RIS, and the receive filters by meeting a sensing SNR constraint. To this end, two DRL algorithms are implemented, and it is numerically demonstrated that the introduced solution outperforms two other related schemes based on traditional RIS approaches.

A RIS-assisted covert ISAC system is investigated in [216] where a radar transmitter aims at detecting a target and covertly communicating with a legitimate user with a low probability of being detected by a warden. To maximize the covert rate under a set of certain restrictions, an optimization problem is cast to design the transmit beamforming vectors and the phase shifts of the RIS. Numerical findings verify that the presented scheme can achieve a notably larger covert rate as compared to an ISAC system with no RIS. The work in [217] considers a RIS-aided ISAC system as well. Here, a dual-functional BS communicates with a single-antenna

communication receiver and senses a malicious eavesdropping target at the same time. In addition, a RIS is exploited to help secure transmission. Assuming that only imperfect CSI of the eavesdropper's channel is available at the transmit side, the transmit beamforming and the RIS reflecting coefficients are optimized to maximize the secrecy rate while meeting certain sensing performance requirements. Numerical results indicate that the introduced algorithm can efficiently manage secure communications even under the adopted imperfect CSI scenario. The use of RIS is promoted to improve PLS for ISAC systems in [218] and [219] where RIS is exploited for both communications and sensing purposes. The scenario of interest encompasses an eavesdropping target and multiple communication receivers served by a common BS. The beamforming vectors for communications and sensing as well as the RIS phase shifts are collectively optimized such that the sensing beam pattern gain is maximized. Both perfect and imperfect CSI cases are separately treated under certain constraints related to secure communications and sensing. Numerical results confirm the strength of the introduced technique in enhancing the PLS of ISAC systems.

In [220], an ISAC BS transmits information to a set of legitimate downlink users while sensing a target and trying to restrain the target from intercepting the communication signal. The system capitalizes on a RIS and designated interference signals formed at the transmitter to augment the secrecy rate as well as to enhance the sensing performance. A RIS-aided ISAC system is studied in [221] by assuming that a multi-antenna BS detects an eavesdropping target and communicates with numerous single-antenna users. AN is added to the aggregate signal at the transmit side to strengthen communication security. Also, active and passive beamforming processes are optimized such that the attainable sum secrecy rate is maximized under a total power limitation and an SINR condition for sensing. The study in [9] includes the integration of RIS into an ISAC scenario where a BS concurrently executes multi-user MISO secure communications and sensing of a malicious target. An optimization problem involving transmit and reflection beamforming and receive filtering processes is devised such that the radar output SNR is maximized under a transmit power limitation as well as communication QoS and security constraints. Numerical results demonstrate that the introduced technique can yield a remarkable advantage in radar performance in comparison to a similar approach without a RIS.

In [222], the beamforming design problem for PLS enhancement is addressed in a RIS-endorsed ISAC scenario involving a multi-antenna BS, a legitimate communication receiver, a target to be sensed, and an eavesdropper. For the sake of a practical design, the BS is assumed to have no CSI knowledge of the eavesdropper's channel. Accordingly, an approximate ergodic achievable secrecy rate expression is derived and used as the objective function to optimize the transmit and receive beamformers, the AN covariance matrix, and the RIS phase shifts under certain communication and sensing performance constraints. Simulation results confirm the validity of the analytical derivation and show the superior performance of the presented scheme as compared to a few other related approaches in terms of the ergodic achievable secrecy rate.

A security-guaranteed ISAC system with an intelligent omni-surface (IOS) is studied in [223]. IOS, a new RIS architecture, is also known as STAR-RIS or STAR-IOS and can accomplish full dimensional coverage using intelligent reflection and refraction of the incident signal toward both sides of the surface at the same time. The sensing beam pattern gain is maximized via a collective design of beamforming vectors and phase-shift matrices under a minimum SINR constraint for communications and a maximum information leakage constraint at the malicious targets. Numerical results demonstrate the potential of IOS technology in enabling secure communications while managing the desired level of sensing performance. In [224], PLS is inspected in an active RIS-aided terahertz ISAC system with a BS, a communication user, and an eavesdropper target to be sensed. By adopting delay alignment modulation at the BS, the secrecy rate is to be maximized while achieving a specific sensing performance level. To this end, transmit beamforming vectors and the active RIS reflection coefficients are optimized. Numerical findings demonstrate the efficacy of the recommended method.

The authors of [225] investigate the secure transmission problem for a RIS-based backscatter communication system where an ISAC BS conveys information to multiple IoT devices and tries to track an aerial eavesdropper to acquire CSI. The sensing outcome is used to construct the transmit waveform and RIS phase shift values such that the sum secrecy rate is maximized while ensuring the required security level and sensing precision. Numerical results prove the potential of the proposed technique in enhancing the sum secrecy rate while achieving desired sensing accuracy levels. A novel transmissive-RIS-based robust and secure ISAC system is developed in [226]. A time-division approach is adopted for JSC, and rate-splitting multiple access is utilized to efficiently manage interference and attain higher spectral efficiency levels. The effectiveness of the introduced architecture beyond other related techniques is demonstrated using numerical results.

B. SECURE ISAC SYSTEMS WITH NOMA

Multiple access techniques and accompanying waveforms have customarily been instrumental throughout the transitions among communications standards. From the first generations to the recent 5G NR, orthogonal multiple access (OMA) approaches have been embraced primarily. The current 5G NR features OFDMA, where orthogonal subsets of time-frequency resources are assigned to distinct users/terminals. OFDMA possesses important advantages such as its potency to deal with frequency-selective fading, its relatively low-complexity implementation, and its ease of

deployment considering backward compatibility. However, OFDMA and other OMA practices can lead to only limited spectrum efficiency as they count on orthogonal resources in either time, frequency, or code domain. In power-domain non-orthogonal multiple access (NOMA), the data multiplexing is performed in the power domain and the receive sides carry out the decoding process by exploiting the power allocation differences among users and/or successive interference cancellation (SIC). Accordingly, NOMA has the potential to offer increased spectral efficiency and user fairness by granting numerous users/terminals to access the same frequency band at the same time.

Multi-user superposition transmission, a downlink NOMA method, has been proposed for the 3GPP LTE advanced (3GPP-LTE-A) networks. Also, it is highly anticipated that NOMA approaches will be adopted in future communications generations. In a NOMA-based ISAC system, multiple nodes make use of the same set of time and frequency resources at different power levels. This power-domain multiplexing can be utilized by attackers to intercept and eavesdrop on the communication and sensing signaling. In addition, the use of power control to uncouple users' signals may result in information leakage about the weaker signals towards unintentional receivers thereby raising serious concerns about user privacy. The shared nature of the communications resources in NOMA-ISAC systems increases the risk of interference and jamming attacks where a wicked entity can intentionally disrupt communication and sensing functionalities by overpowering weaker signals to misalign signals or by jamming the whole used spectrum. Intending to fulfill the demanding challenges of next-generation wireless networks securely and robustly, several studies have considered the integration of NOMA schemes into ISAC scenarios from a security point of view.

In [64] and [227], the advantages of NOMA are reaped to serve an increased number of users over an ISAC network by maintaining a certain degree of security. A BS transmits secure information to a set of users and senses the position of a target at the same time. The target is assumed to possibly intend to wiretap the downlink communications. The transmitted superimposed signal at the BS is determined such that the sum secrecy rate is maximized for multiple users meanwhile meeting a certain level of sensing performance. Numerical findings indicate the superior performance of the presented system as compared to OMA-ISAC. The work in [228] considers an active RIS-aided NOMA-ISAC system composed of a BS with an antenna array, an active RIS, multiple targets with LoS links to the BS, a warden acting as a malicious surveillant to detect the signal transmission, a public far user, and a covert near user. To maximize the covert rate while satisfying the public user's QoS demand, a certain sensing performance, and a definite covertness grade towards the warden, the transmission and reflection beamforming processes are optimized. Simulation results exhibit that the adopted active RIS-aided NOMA-ISAC system performs better than its counterpart with a passive RIS concerning the covert rate and sensing beam pattern similarity metrics.

In [229], it is speculated that the preceding NOMA-ISAC systems may be unsuccessful in ensuring transmission security when some of the legitimate receivers are located at similar angles regarding BS. To tackle with this problem, a beamforming and power allocation policy is devised based on user clustering. The considered system model includes imperfect CSI and SIC cases as well. Simulation results show that the presented scheme provides a high detection probability of targets while supporting a desired level of security rate. A NOMA-ISAC scenario is covered in [230] by using STAR-RIS to form LoS channels for sensing a target and conveying confidential messages to multiple users. To maximize the sum secrecy rate, the transmit beamforming, the artificial jamming, and the transmission and reflection vector of the STAR-RIS are optimized. Numerical results show the advantage of the proposed system in terms of secure communications and sensing ability. In [231], a NOMA-aided secure transmission scheme is investigated for RIS-ISAC systems. Here, a BS communicates with multiple users via NOMA and senses a target which also behaves as a suspicious eavesdropper. To maximize the sum secrecy rate, a collective precoding and reflecting optimization problem is drafted by forcing that a certain sensing performance is achieved. Numerical findings indicate that the presented technique notably outperforms the other related benchmark schemes.

C. SECURE ISAC SYSTEMS: USE CASES WITH UAV

Due to their salient traits such as maneuverability, adjustable altitude, cost-effectiveness, and adaptability to distinct scenarios, UAVs are expected to play core roles in enabling ubiquitous, seamless, and robust connectivity as well as improved coverage in the upcoming communication systems. To make use of the advantages of two versatile technologies, UAV-enabled ISAC systems have recently drawn a great deal of attention from different perspectives. In the following, a survey of UAV-enabled ISAC systems with a security viewpoint is provided.

In [232], the authors discuss ISAC-enabled UAV communications by providing motivation cases and distinct viewpoints such as security, latency, and reliability requirements. It is stated that due to the common use of the spectrum and the intrinsic propagation attribute of the wireless channel, UAV terminals are open to Sybil attacks in which a legitimate node is hacked and the attacker allegedly asserts that it possesses various imaginary identities. Such a situation manipulates other legitimate terminals to mistakenly perceive that additional nodes are added to the system. This may result in miscellaneous disruption as well as security and efficiency issues. Driven by this problem, a novel ISAC-enabled dual identity solution is introduced for trusted and swift UAV communications. Three typical application scenarios are presented together with the benefits of the recommended ISAC-assisted dual identity solution in each case. A secure

UAV-assisted dynamic ISAC system is inspected in [233] where the UAV acts as a downlink transmitter and a radar receiver. It is assumed that a legitimate user wanders on the ground through a sequence of anonymous spots and an eavesdropper proceeds through a given trajectory. The UAV tracks the legitimate user to enhance the performance of the communications. A weighted real-time trajectory design problem is devised to maximize the real-time secrecy rate. The ensuing non-convex optimization problem is solved using an iterative algorithm to yield a near-optimal solution. Numerical findings suggest that the introduced system model concurrently enables both resilient secure communication and precise tracking.

The work in [234] inspects a UAV-aided ISAC network with a dual-functional BS, a UAV used as a relay, a legitimate user, and an active eavesdropper both on the ground. The BS senses the eavesdropper's position and transmits information to the legitimate user. A Stackelberg game model is introduced for secure information transfer considering both perfect and imperfect CSI cases. A RIS-assisted UAV-ISAC scenario is covered in [235] with a UAV BS communicating with multiple users and sensing various targets at the same time through the help of a RIS. In addition, a prospective eavesdropper intends to maliciously listen to the transmission between the BS and the users. Assuming no eavesdropper CSI at the transmitter, a secure communications method is recommended to maximize the average attainable rate through collectively designing the transmit power allocation, the phase shifts at RIS, and scheduling together with the trajectory and speed of the UAV.

New security measures are introduced in [236] to tackle the jamming and eavesdropping attempts in RIS-UAV-enabled ISAC systems. For the jamming attacks, a RIS-UAV executes passive beamforming meanwhile bridging a LoS connection for the sensing process of a blocked target. In addition, the RIS-UAV governs AN to guarantee the precision of target detection meanwhile eliminating eavesdropping. The presented numerical results show the potency of the introduced techniques. The study in [237] investigates a low earth orbit satellite-terrestrial integrated network by adopting NOMA to enable enhanced coverage and service. Depending on the condition of the CSI for the eavesdropper's channel at the transmitter, separate precoding, and AN optimization problems are treated to maximize the sum secrecy rate for multiple users at the same time while meeting certain QoS constraints such as the echo SNR and the sum transmit power. Numerical findings demonstrate that the presented scheme provides an improved sum secrecy rate as compared to a conventional approach based on time-division multiple access.

While many of the reviewed techniques, such as joint beamforming and AN strategies, effectively enhance secrecy performance, they often rely on idealized assumptions such as perfect CSI or static environments, which may limit their practical deployment. Furthermore, the computational complexity associated with some optimization schemes could hinder real-time implementation in fast-varying scenarios. To overcome these limitations, future research could explore adaptive ML-based methods that dynamically optimize security parameters with reduced complexity. Additionally, integrating cryptographic techniques with PLS functionality could provide more comprehensive protection, especially against sophisticated eavesdroppers. Experimental validation using hardware prototypes or field trials is also essential to evaluate the real-world viability of these approaches.

VI. AI/ML IN ISAC

ISAC is becoming a key paradigm in 6G communication platforms. It can utilize the complementary functions of sensing and communication to improve situational awareness and operational efficiency. Advanced signal processing and decision-making capabilities are made possible by DL, which is essential to this integration due to its exceptional capacity to extract meaningful patterns from datasets. ISAC systems can optimize sensing and communication processes by dynamically adapting to complex and rapidly changing surroundings through the utilization of ML methods. This part of the literature review examines the relationship between ISAC and ML algorithms, emphasizing how their confluence could revolutionize the field and advance the state-of-the-art in various tasks.

A. ML TRENDS IN ISAC

ML has emerged as a transformative force within this field, driving innovation and efficiency in ISAC systems. Current trends in ML for ISAC focus on enhancing spectrum utilization, improving signal processing, and optimizing resource allocation. These advancements promise applications in autonomous vehicles, smart cities, and beyond, by providing robust solutions to the complexities of dynamic and interconnected environments.

The intersection of ISAC with ML to enhance future wireless systems is comprehensively explored in [238]. The study highlights the importance of ISAC, which aims to merge communication and sensing functionalities to optimize spectrum usage and improve overall system efficiency. It emphasizes ten key roles of ML in this context, including signal processing, resource allocation, and network management. These roles demonstrate how ML can address the complexities and challenges posed by the ISAC in 6G networks, offering potential solutions through advanced algorithms and data-driven approaches. Table 3 covers several topics that consider ML-ISAC cooperation.

The first emerging area has been JSC, which involves the simultaneous use of communication and sensing functions, sharing the same hardware, waveform, and wireless resources. This integration aims to maximize the efficiency of wireless systems by leveraging the capabilities of both functions. However, it also presents several challenges that can be addressed through the use of ML. One of the primary roles of data-driven systems in JSC is the optimization of waveform design. Creating a single waveform that effectively

TABLE 3. Several topics that consider ML-ISAC cooperation [238].

Category	Role
Joint Sensing and Communication	1. Waveform optimization 2. Learning spatial beam patterns 3. Self-interference cancellation for full-duplex 4. Resource optimization 5. Enhancing system security 6. Enabling network operation
Sensing Aided Communication	7. Sensing-aided beam prediction 8. Sensing aided blockage and handover prediction
Communication Aided Sensing	9. Communication-aided sensing optimization 10. Communication-aided network sensing

serves both radar and communication functions requires balancing trade-offs between the two operations, such as data rates, latency, reliability, and sensing capabilities. AI models can help design these joint waveforms to meet the objectives of both functions efficiently. Another significant role is learning site-specific spatial sensing and communication beam patterns. This involves proper beamforming patterns that suit specific environments, thereby improving the accuracy and efficiency of both communication and sensing.

Another significant topic handled in [238] is sensing-aided communication (SAC). This approach uses sensory data to enhance wireless communication operations. For example, in systems with large antenna arrays, sensing can facilitate beam prediction and tracking. In millimeter wave and sub-terahertz systems, where LoS link blockages pose significant challenges, sensing can aid in blockage prediction and proactive handover. This predictive capability allows the network to make proactive decisions on handover and beam switching, improving reliability and reducing latency. ML can be used in SAC by enabling these predictive and adaptive functionalities through the analysis of sensory data and integration with communication systems.

Additionally, AI enables sensing-aided blockage prediction and proactive handover, addressing critical challenges for the reliability and latency of the network in high-frequency bands. Communication-aided sensing (CAS) involves using existing communication signals, such as transmitted preambles, for sensing purposes. The communication signals transmitted to a receiver can also reflect off various objects in the environment, allowing for tasks such as positioning, object detection, tracking, and imaging. This approach is passive and does not interfere with the performance of the communication system since the sensing objective is not part of the original signal design. CAS is valuable in scenarios where communication is prioritized or where the same hardware is used for both communication and opportunistic sensing. Learning-based systems help to optimize CAS by processing received communication signals to achieve high sensing quality. Traditional signal processing approaches often struggle with clutter noise, multipath reflections, and unoptimized side lobes. However, ML can leverage data-driven learning to optimize signal processing without explicit modeling of hardware or channel effects.

The dual role of UAVs in providing both communication and sensing capabilities can be utilized in various applications like surveillance, disaster response, trajectory design, and environmental monitoring are described in [233], [239], and [240]. Key areas discussed include resource allocation, trajectory planning, and interference management. In [241], UAV-enabled ISAC systems use AI to enhance UAVs' performance by managing tasks like beamforming, interference mitigation, and real-time decision-making. Through ML algorithms, UAVs can dynamically adjust their flight paths and resource use to balance the needs of communication and sensing tasks, considering factors like energy efficiency and signal quality.

The transformative potential of generative artificial intelligence (GAI) in enhancing the performance and efficiency of ISAC systems are explored in [242], [243], and [244]. They thoroughly examine various GAI models, including generative adversarial networks (GANs), variational autoencoders, and diffusion models, highlighting their unique abilities to generate, simulate, and analyze data. These capabilities are particularly significant for improving physical layer technologies, crucial for the simultaneous execution of communication and sensing functions. In [244], the authors explore the application of GAI in crucial tasks such as channel estimation, signal detection, and waveform design and demonstrate that these models can significantly enhance the accuracy and robustness of ISAC systems. A case study focusing on the estimation of the direction of arrival of signals in near-field conditions underscores the practical benefits of GAI, where an MSE of 1.03 degrees squared is achieved, showcasing the precision and reliability of these AI-driven approaches.

References [160], [245], [246], and [247] examine AI integration at the wireless network edge, a key step for 6G development, focusing on edge intelligence for low-latency and real-time processing. They propose an ISAC framework that meets 6G's demands for ultra-low latency, high data rates, and massive connectivity by optimizing resource use, data processing, and response times, essential for AI-driven applications like autonomous vehicles, smart cities, and the metaverse. Reference [245] presents an ISAC-oriented IoT architecture model and reviews ISAC technology development. It outlines new design principles for AI-empowered, multimodal networks, addresses security issues in ISAC, and inspects various layer architectures and ML methods. It also discusses current technological limitations and future research directions. Reference [247] focuses on AI/ML integration in the near-field ISAC systems for next-generation networks, using DL to optimize hybrid beamforming and predictive beam tracking. The study shows that RIS can enhance ISAC performance through virtual LoS pathways and passive beamforming, and it explores networked sensing with multiple BSs using ML for improved radar sensing precision.

In this section we provide an overview of the trends in utilizing AI and ML models within ISAC systems, highlighting their growing significance and potential benefits. The subsequent parts cover a more technical discussion, focusing

on how these models are integrated and operate within ISAC frameworks.

B. DEEP NEURAL NETWORK BASED APPROACHES

The integration of communication and sensing functions within modern autonomous systems presents significant challenges and opportunities for enhancing operational efficiency and performance. DNNs, with their ability to model complex, non-linear relationships and learn from vast amounts of data, have become a cornerstone of contemporary solutions in this domain. DNN-based approaches offer advanced capabilities for optimizing the coordination and resource management of ISAC systems. These techniques enable precise signal processing, robust environmental perception, and efficient spectrum utilization, thereby addressing the critical demands of dynamic and densely populated scenarios.

In ISAC systems, topics such as channel estimation, enhanced sensing capabilities, and waveform design are crucial for optimizing both communication and sensing functions. Especially, the use of AI models in these areas plays a vital role in solving non-convex problems, accelerating the system, or finding better effective solutions [13], [65], [66], [248], [249]. Reference [13] proposes a three-stage channel estimation method for RIS-assisted ISAC systems to address interference between sensing and communication signals. By using convolutional neural networks (CNNs) at each stage, the approach significantly reduces MSE across varying SNR levels, outperforming traditional methods. Another study, [65], introduces ISAC-NET, a model-driven DL framework that integrates passive sensing and communication, improving spectral efficiency and overall performance. This framework leverages shared resources to overcome challenges in signal processing without prior knowledge of transmitted signals, enhancing accuracy and security. Additionally, [66] focuses on designing sensing waveforms and communication beamforming using a DL-based network, ISACNN, to address interference at the BS. By transforming a complex non-convex optimization problem into a solvable framework, the study achieves efficient waveform design under power constraints. The general framework of the study is given in Fig. 11.

Vehicular networks and communication are of paramount importance, particularly as the automotive industry moves towards autonomous and connected vehicles. These networks facilitate real-time data exchange between vehicles and infrastructure, enabling precise situational awareness and decision-making. ISAC systems leverage DNNs for beamforming design, target tracking, and improving the efficiency [67], [250], [251], [252], [253], [254]. Reference [67] introduces an end-to-end predictive beamforming (E2E-PB) method for V2I networks, using an attention-based LSTM network to directly map signal samples to beamforming vectors without intermediate parameter estimation. This unsupervised approach maximizes communication rates and outperforms traditional methods like extended Kalman filtering, achieving near-optimal beamforming performance under ideal conditions. Reference [253] proposes a novel ISAC framework employing OFDM and NOMA to address challenges in multi-user demodulation and target tracking. CNN-based schemes treat data reception as a classification problem, improving accuracy and reducing errors. A YOLOv5-SORT scheme integrates radar signal processing with computer vision for precise multi-target detection, achieving high performance in terms of symbol error rates and F1 scores. Reference [254] explores DL-based predictive beamforming to optimize ISAC in V2I networks. The proposed HCL-Net model leverages spatial and temporal dependencies in historical channel data to improve beamforming matrix predictions. This approach addresses training overhead and computational challenges, maximizing achievable sum rates while meeting sensing constraints.

In ISAC networks, there are scenarios where multiple devices communicate with each other. Different strategies are required for training the large volumes of data generated in these situations. Federated learning (FL) allows models to be trained across multiple devices without centralizing the data, which is crucial for privacy-sensitive environments. This method is particularly useful in scenarios like ISAC, where data from different sources must be processed locally. In contrast, DL typically requires large, centralized datasets to train complex models, making it less suitable for decentralized environments like those in ISAC [255], [256], [257]. Reference [258] employs FL to optimize resource allocation in ISAC networks, introducing a multi-objective optimization model that balances sensing precision, power consumption, and communication rates. The model is tackled using multi-task learning with DL techniques like the multiple-gradient descent algorithm. The study highlights the evolution of ISAC from resource-sharing coexistence to full integration, where ML enhances both hardware integration and protocol development to meet the complex demands of 6G, such as high data rates, low latency, and massive connectivity. In [144], the authors propose the FedFog architecture, a three-layer structure comprising terminal perception, edge BS processing, and cloud data center layers. This architecture leverages edge intelligence and FL to process data locally, reducing latency and addressing privacy concerns by minimizing raw data transmission.

In the training of DL methods, supervised and unsupervised learning are widely used. The papers presented under this subsection primarily cover models developed using the specified learning techniques. Another learning method frequently employed, especially in the training of robots, is DRL. In this strategy, the model attempts to optimize a given cost function based on the environment. Hence, RL is particularly utilized in dynamic applications such as planning, trajectory extraction, and resource optimization. This approach allows the model to learn optimal behaviors through interaction with its environment, making it suitable for tasks requiring adaptability and decision-making.

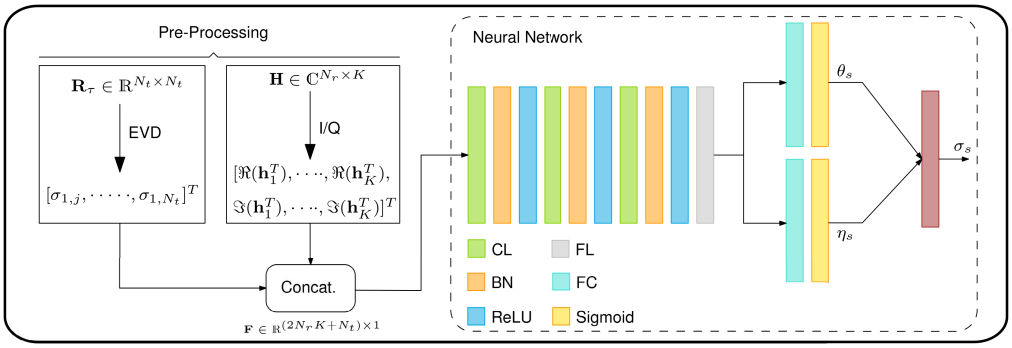


FIGURE 11. General structure of ISACNN architecture. $\mathbf{R}_T$ and H are sensing transmit covariance matrix and communication channel matrix, respectively. Their combination is the input vector of the proposed DNN model. The aim of the architecture is an estimation of the transmit waveform's singular values σ_S.

C. REINFORCEMENT LEARNING BASED APPROACHES

The rapid advancement of autonomous systems has necessitated the development of sophisticated techniques for optimizing their performance, particularly in the realms of ISAC. DRL has emerged as a powerful tool for addressing complex decision-making problems in dynamic environments. By enabling systems to learn optimal policies through interactions with their surroundings, RL-based approaches have shown great promise in enhancing the efficiency and effectiveness of ISAC systems.

The integration of MIMO, RIS, and DRL significantly enhances the performance of ISAC systems, especially in 6G networks, by enabling efficient resource management, improved beamforming, and enhanced sensing accuracy [68], [259], [260], [261]. A RIS-aided simultaneous localization and communication (SLAC) framework in cloud radio access networks (C-RAN) is introduced in [68]. This framework uses hybrid DRL, specifically the deep deterministic policy gradient algorithm, to optimize RIS phase shifts, subcarrier assignments, and beamforming. It improves resource efficiency and localization accuracy while addressing the computational challenges of high-dimensional decision spaces. To tackle multi-user MIMO downlink scenarios, [69] proposes a 2D fluid antenna system that optimizes antenna port positions and precoding. Using DRL methods like primal-dual learning and an advantage actor-critic framework, the study solves the NP-hard optimization problem and enhances user sum rates under sensing constraints. For autonomous vehicles, [262] presents a time-division ISAC system that shares sensing and communication resources. A DRL-based algorithm optimizes time-frequency resource allocation, reducing interference and energy consumption. By employing a partially observable Markov decision process, the study models multi-vehicle signal interference, improving detection accuracy and resource management in dynamic environments.

Advances in 5G and IoT technologies have increased the number of sensor nodes, leading to challenges in energy and time efficiency due to higher demands for computation and communication resources. To address this, Cai et al. propose a deep deterministic offloading and resource scheduling framework, a DRL-based technique aimed at minimizing latency and energy consumption by optimizing the offloading of computational tasks from edge devices (EDs) to edge computing (EC) servers [157]. The problem, formulated as a mixed-integer nonlinear programming task, is split into two subproblems. The first, focusing on local computation, is solved using convex optimization, while the second, addressing offloading and resource allocation, uses a double-deep Q-learning network for decision-making. The framework models a system with a BS equipped with EC servers and multiple EDs, each characterized by task parameters like input data size, CPU requirements, delay tolerance, and battery capacity. The optimization seeks to minimize the combined delay and energy consumption while adhering to constraints on task completion time, energy usage, and resource capacities. This approach efficiently balances computational loads between EDs and EC servers, offering a scalable solution to resource allocation challenges in 5G and IoT networks.

Another DRL-based study focuses on enhancing security in ISAC systems using RIS [12]. It addresses the challenge of secure communication in scenarios where a BS interacts with multiple users while tracking a potential eavesdropper. The proposed approach maximizes the secrecy rate for legitimate users by jointly optimizing transmit beamforming, AN signals, and RIS phase-shift settings. Due to the problem's complexity, a soft actor-critic method is used, which improves policy diversity and avoids local optima by maximizing policy entropy. Simulations demonstrate that the soft actor-critic-based approach outperforms traditional methods, such as deep deterministic policy gradient, and systems without AN or with random RIS phase shifts, significantly enhancing the secrecy rate. Reference [263] proposes a cooperative sensing and RIS-based approach to enhance UAV path-following and obstacle avoidance, crucial for applications like cargo transport and rescue missions. The study introduces a DRL-based model with adjustable reward function weights to minimize velocity and distance errors in UAV formation path-following. Obstacles are detected using

ISAC signals, and a variable formation-enhanced algorithm improves obstacle position accuracy using information-level fusion and CRLB calculations. An online obstacle avoidance scheme, designed without pre-training, addresses the sparse reward problem, while a hierarchical subtasks fusion strategy ensures conflict-free integration of path-following, sensing, and avoidance tasks. Simulations confirm the effectiveness of the proposed methods in improving UAV formation control and safety.

ISAC platforms frequently rely on DNNs, RL, and FL, each bringing distinct advantages and limitations. DNNs are particularly effective at capturing complex, nonlinear relationships and have demonstrated state-of-the-art performance in tasks such as target detection, channel estimation, and joint waveform optimization without requiring explicit signal models. However, their dependency on large volumes of labeled training data and substantial computational resources can pose significant challenges for on-device training and real-time adaptation in resource-constrained environments. RL methods stands out in dynamic scenarios—such as vehicle-to-infrastructure networks or unmanned aerial vehicle operations—where channel conditions and interference patterns vary rapidly. By interacting with the environment and optimizing long-term rewards, RL agents (e.g., those based on Soft Actor-Critic) can adapt beamforming and resource allocation policies on the fly, yielding robust performance under nonstationary conditions. Nonetheless, RL's typically slow convergence rates and the difficulty of transferring policies from simulation to real-world deployments remain practical hurdles that must be carefully managed. FL offers a compelling alternative when data privacy or limited bandwidth preclude centralized data aggregation. In federated schemes, edge nodes collaboratively train a shared model by exchanging only weight updates instead of raw sensor data, thereby preserving privacy and reducing communication overhead in large-scale IoT deployments. On the other hand, this approach may face challenges such as heterogeneous client capabilities, synchronization delays, and security vulnerabilities like model poisoning to ensure reliable convergence and robustness.

Although DNNs can learn critical patterns from high-dimensional radar–communication measurements, they tend to depend on large labeled datasets and are propagation-variability sensitive. In order to mitigate these challenges, the application of physics-informed regularization to the training objective and the exploration of semi-supervised methods to harness unlabeled over-the-air data are recommended. Another problem to be faced in ISAC networks, an obstacle to FL is posed by the untrusted nature of wireless channels, which tend to delay, corrupt, or even lose model updates. Low signal-to-noise ratios, packet loss, and fluctuating latency hinder the timely delivery of local updates to the aggregation server. Hence, this scenario adds bias to the global model and delays convergence. To alleviate this issue, a dynamic client selection and weighting strategy can be suggested. Channel quality is estimated by a brief pilot transmission prior to each federated round. Then, the updates of clients with reliable transmission are given higher aggregation weight, whereas those of nodes with poor channel conditions are deferred to future rounds. By prioritizing transmissions from robust links and postponing unreliable transmissions, communication overhead along with global convergence time can be significantly reduced.

In this section, the impact of AI/ML models on ISAC systems is handled across various tasks. By leveraging advanced neural network architectures such as DNNs, ISAC systems can achieve unprecedented levels of performance in both communication and sensing tasks. In conclusion, the adoption of DL architecture in ISAC communication systems significantly advances the capabilities and effectiveness of modern wireless communication, paving the way for future innovations in intelligent transportation and beyond.

VII. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

This survey provides a comprehensive overview of ISAC systems, examining them from both sensing-centric and communication-centric perspectives. We have explored innovative approaches to enable ISAC, including waveform design, edge computing, and vehicular communication, along with their practical implementations. The integration of sensing and communication functions presents significant opportunities to enhance system efficiency and performance, particularly in the context of evolving technologies such as 5G and beyond. Our findings underscore the importance of secure ISAC systems and highlight areas for further investigation, including RIS, NOMA, and UAVs. Additionally, we review the pivotal role of AI/ML in facilitating advancements in new ISAC systems.

Looking towards the future, continued exploration of innovative approaches and technologies is essential for realizing the vision of next-generation communication networks and advancing ISAC capabilities. The growing interest in distributed MIMO and cooperative systems reveals the expanding implications of distributed sensing as a fruitful area for future research. While NLoS performance increases in terms of sensing, challenges related to synchronization and processing arise, highlighting the need for targeted attention.

Another critical area that warrants further investigation is channel modeling specific to ISAC systems. While established channel models exist for communication systems, there is a need to develop models that effectively integrate communication and sensing functionalities. Furthermore, enhancing communication efficiency through sensing techniques by accurately determining the locations of communication terminals may yield substantial benefits.

In addition to theoretical advancements, recent studies have increasingly emphasized practical implementations and real-world applications of ISAC technologies. Prominent areas such as vehicular networks, UAV communication systems, industrial automation, and healthcare stand to gain significantly from the synergy between communication

and sensing. Numerous experimental studies and testbed implementations have been reported, demonstrating the feasibility of ISAC technologies, particularly in scenarios involving beamforming, localization, and radar-assisted communication.

Moreover, there has been growing interest in standardization efforts related to ISAC. 3GPP has initiated studies on ISAC use cases in Release 18, continued developments in Release 19, and is expected to further expand ISAC capabilities in Release 20. These developments indicate that future research should encompass not only theoretical advancements but also practical considerations, including experimental validation, hardware design, and ongoing standardization processes, to effectively support the realization of ISAC systems.

GLOSSARY

- **3CLS** is an abbreviation for communications, computing, control, localization, and sensing.
- **3GPP** is a worldwide consortium that develops technical specifications for mobile communication systems, including 4G, 5G, and upcoming 6G standards.
- **CFAR** is a detection technique that uses adaptive thresholding to find peaks in radar systems under a fixed false alarm probability, independent of background noise fluctuations.
- **CRLB** provides a theoretical lower bound on the variance of unbiased estimators in statistical estimation theory.
- **DRL** combines reinforcement learning with deep neural networks, enabling agents to learn decision-making policies by interacting with environments and optimizing cumulative rewards. Its ability to process high-dimensional inputs makes it suitable for complex tasks such as robotics, autonomous driving, and strategy games.
- **Digital Twin** is a real-time digital replica of a physical system or process. It allows for continuous monitoring, simulation, and optimization, and plays a critical role in smart manufacturing, autonomous systems, and predictive maintenance.
- **FL** is a decentralized machine learning method where multiple devices collaboratively train a shared model without sharing local data. Only model updates are sent to a central server, enhancing privacy, security, and efficiency—especially useful for edge devices, IoT, and sensitive fields like healthcare.
- **Fluid Antennas** uses a reconfigurable liquid or fluid medium to dynamically change its shape or position.
- **GAI** refers to an advanced form of artificial intelligence characterized by the ability to autonomously perform a wide range of cognitive tasks across diverse domains, demonstrating adaptability, reasoning, and learning capabilities beyond narrow or domain-specific applications.
- **ISAC** is an integrated system that has communication functions as well as sensing features. It is seen as an important technology for 6G systems by 3GPP.
- **ISAR** is a radar imaging technique that uses the relative motion between the target and the radar to generate high-resolution images.
- **MEC** brings computation and storage resources closer to end-users by deploying them at the edge of the network. This architecture reduces latency and enhances the performance of latency-sensitive applications such as autonomous driving and immersive AR/VR.
- **mmWave** refers to the frequency band ranging from 30 GHz to 300 GHz. It offers extremely high data speeds and wide bandwidth.
- **NOMA** is a multiple access technique that allows multiple users to share the same frequency-time resources by using power domain separation.
- **PAPR** is a metric used to measure the ratio between a signal's peak power and average power, especially in multicarrier systems like OFDM. High PAPR is a crucial problem.
- **RIS** consists of programmable surfaces that can manipulate electromagnetic waves intelligently to increase SNR/ SINR values.
- **Stealth Technology** involves design methods and materials that reduce the detectability of objects, especially vehicles, by radar, infrared, and other detection methods.
- **V2X** encompasses various forms of communication between a vehicle and its environment, including V2V, V2I, and V2P.

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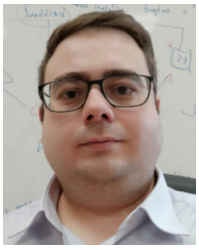
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